Monetary Policy

CHAPTER 15 · PRINCIPLES OF MACROECONOMICS

Money demand, interest rates, and aggregate demand

November 3, 2026

Guilherme Paiva Pinto · ECON_OX 112

Two FOMC Decisions

Earlier decision cards	June 15, 2022 · Case C	September 18, 2024 · Case E
Change in target range	Increase of 0.75 percentage point	Decrease of 0.50 percentage point
New target range	1.50–1.75%	4.75–5.00%

- The 2024 cut left the target range above its level after the 2022 increase.

FOMC, June 15, 2022 · FOMC, September 18, 2024

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- The 2024 cut left the target range above its level after the 2022 increase.
- How could either decision change the money people hold and the purchases firms make?

FOMC, June 15, 2022 · FOMC, September 18, 2024

Learning Objectives

- A · Understand and explain demand for money.
- B · Explain money-market equilibrium and the connection between money and interest rates.
- C · Interpret aggregate-demand shifts and demonstrate the effects of monetary policy.

15.1 Demand for Money

- Understand money demand as a desired stock for payments and liquidity.
- Explain how an investment’s return changes the opportunity cost of holding money.
- Distinguish a movement along money demand from a shift caused by other conditions.

Money for Bills or Money for a Return?

- Use the separate PollEv activity: P1–P4. All amounts, rates, and contracts are hypothetical.
- Choose for the fictional household. Keep enough money for its upcoming bills.
- Keep your reasoning: we will use the first responses to build a money-demand comparison.

The Opportunity Cost of Holding Money

Holding $600 for one year	Ending balance	Interest forgone by holding money
Cash/checking at 0%	$600	—
One-year bond at 1%	$606	$6
One-year bond at 6%	$636	$36

- Rate of return = (ending value − initial value) ÷ initial value.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

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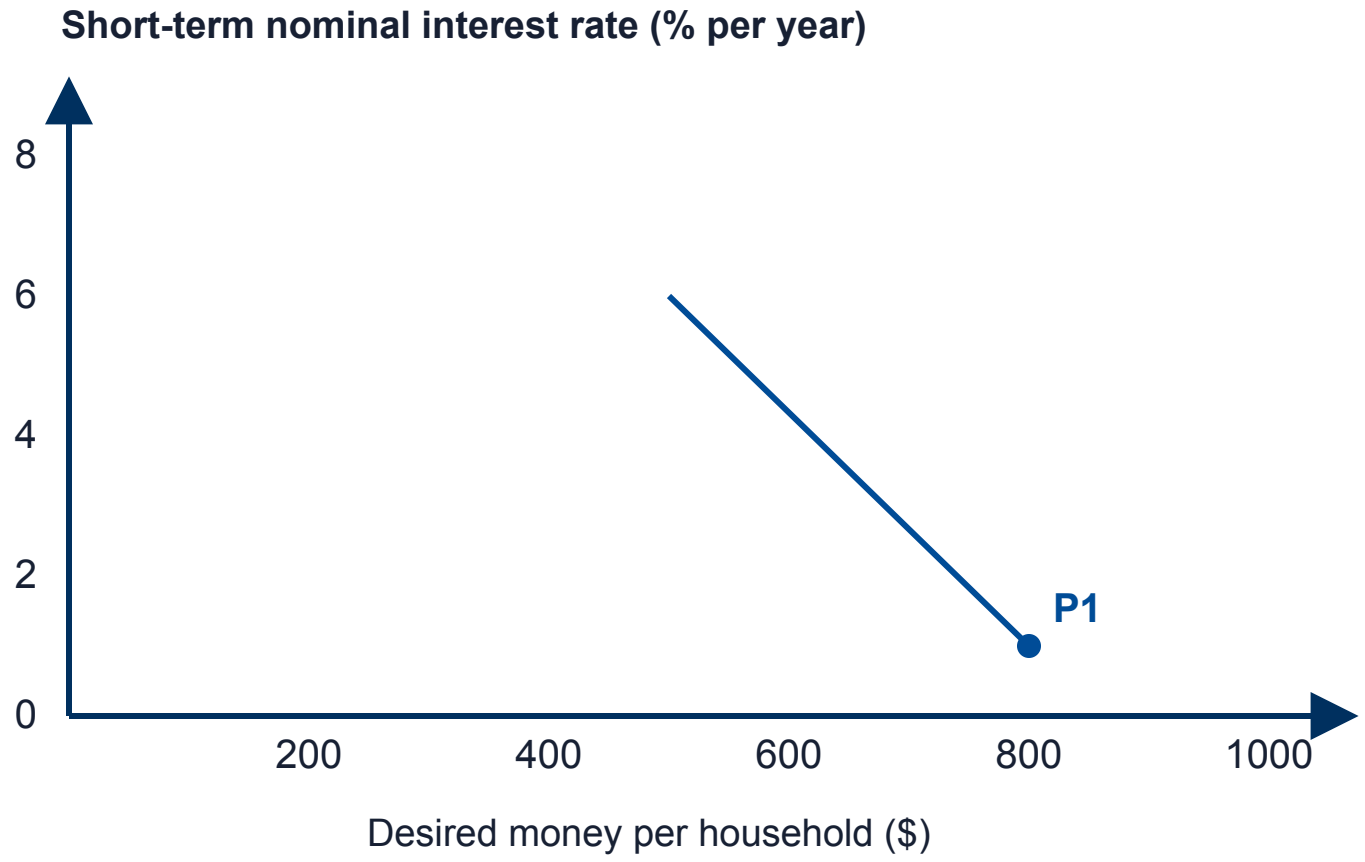
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- Compare with the return paid on money itself; today the activity holds that return at 0%.

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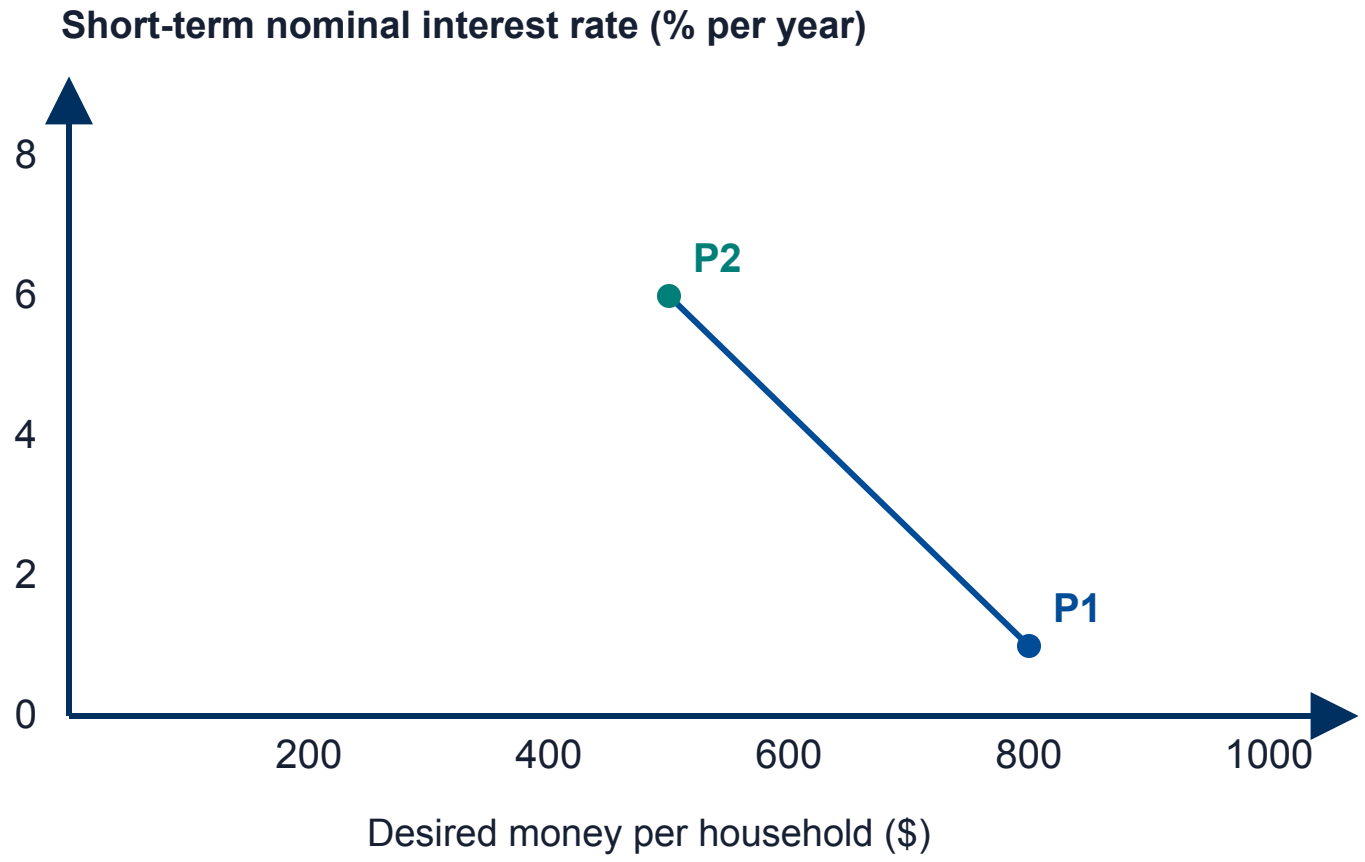
From Choices to Money Demand



Schematic illustration; positions are model examples, not observed class responses.

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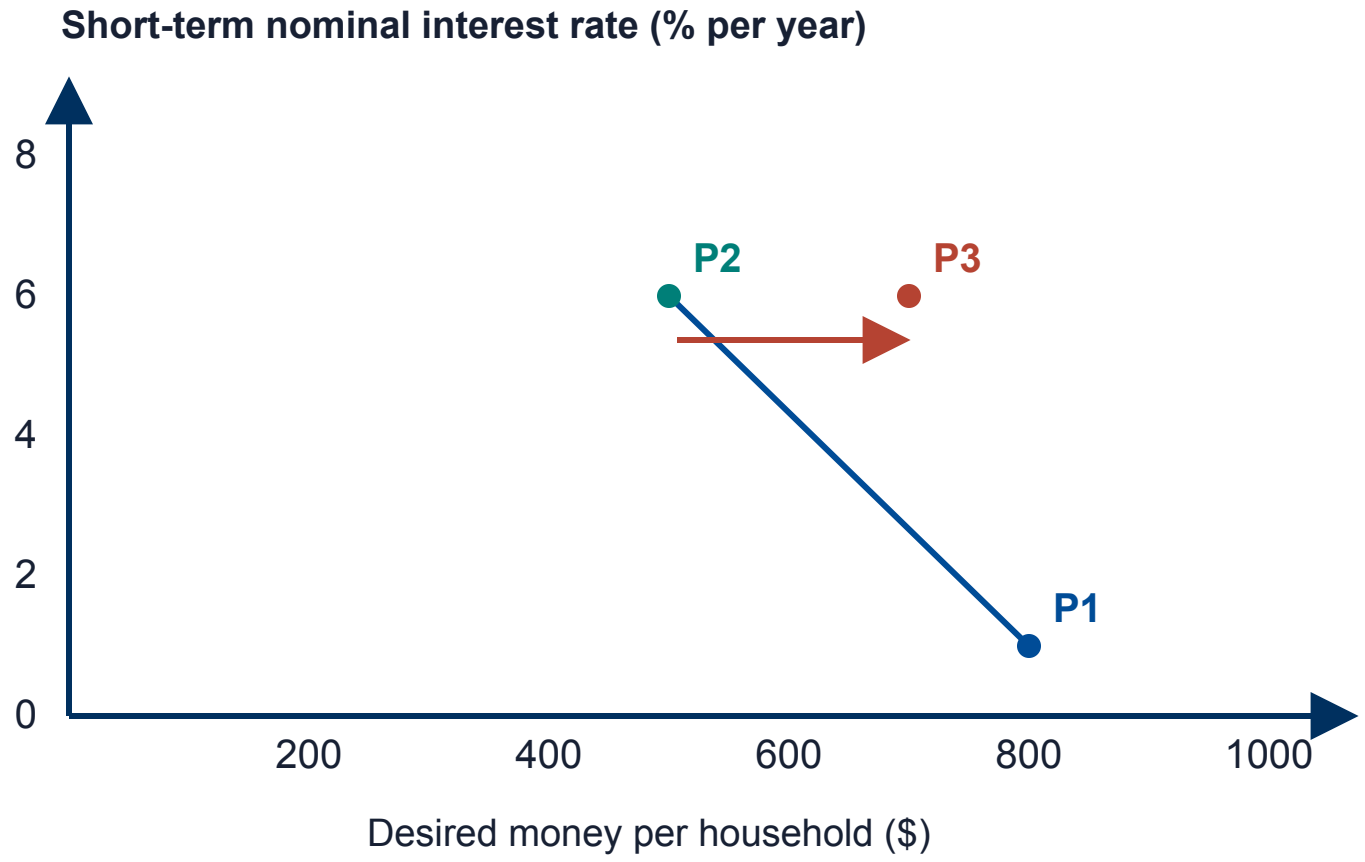
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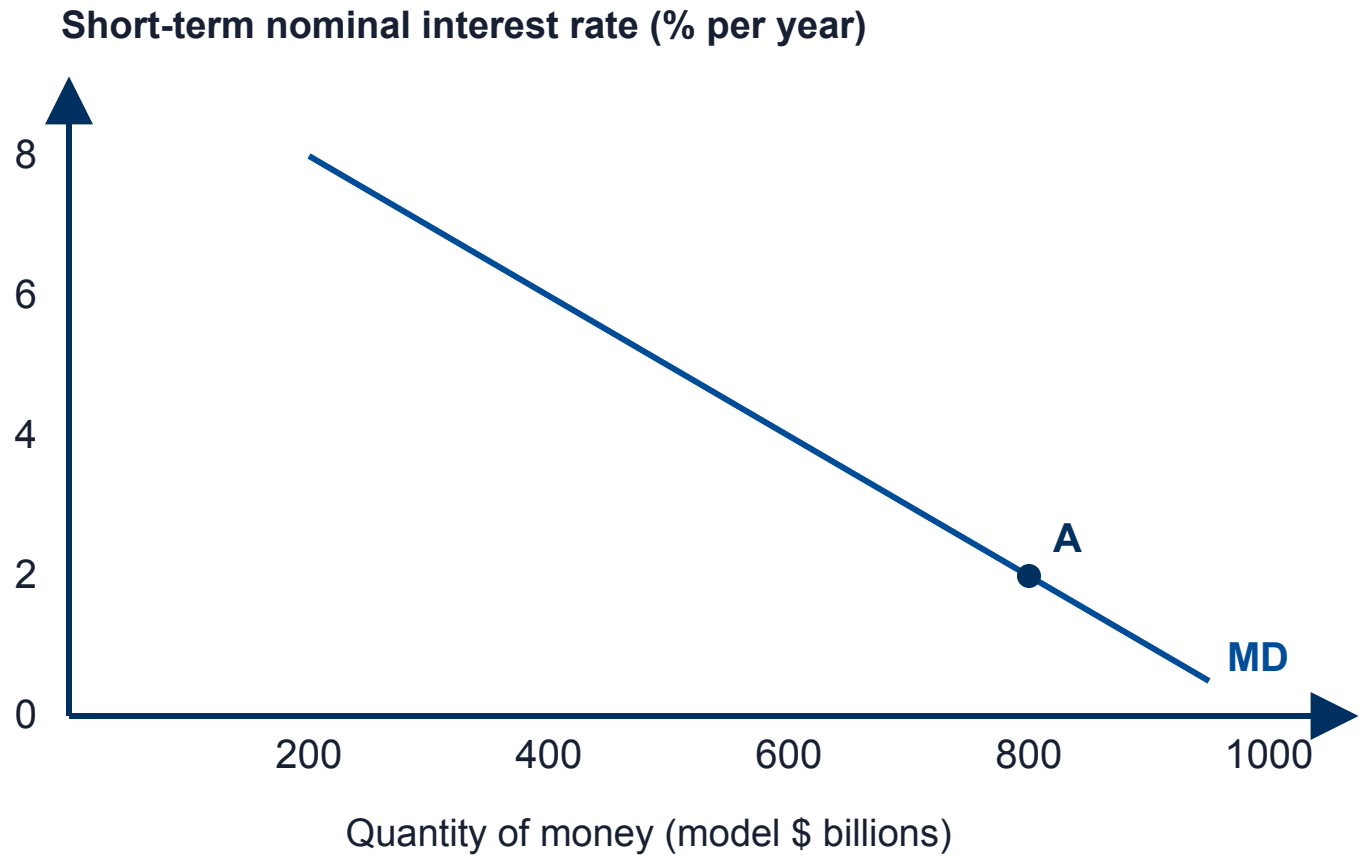
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3. P2 → P3: bills become more expensive at the same 6% rate. That is a demand shift.

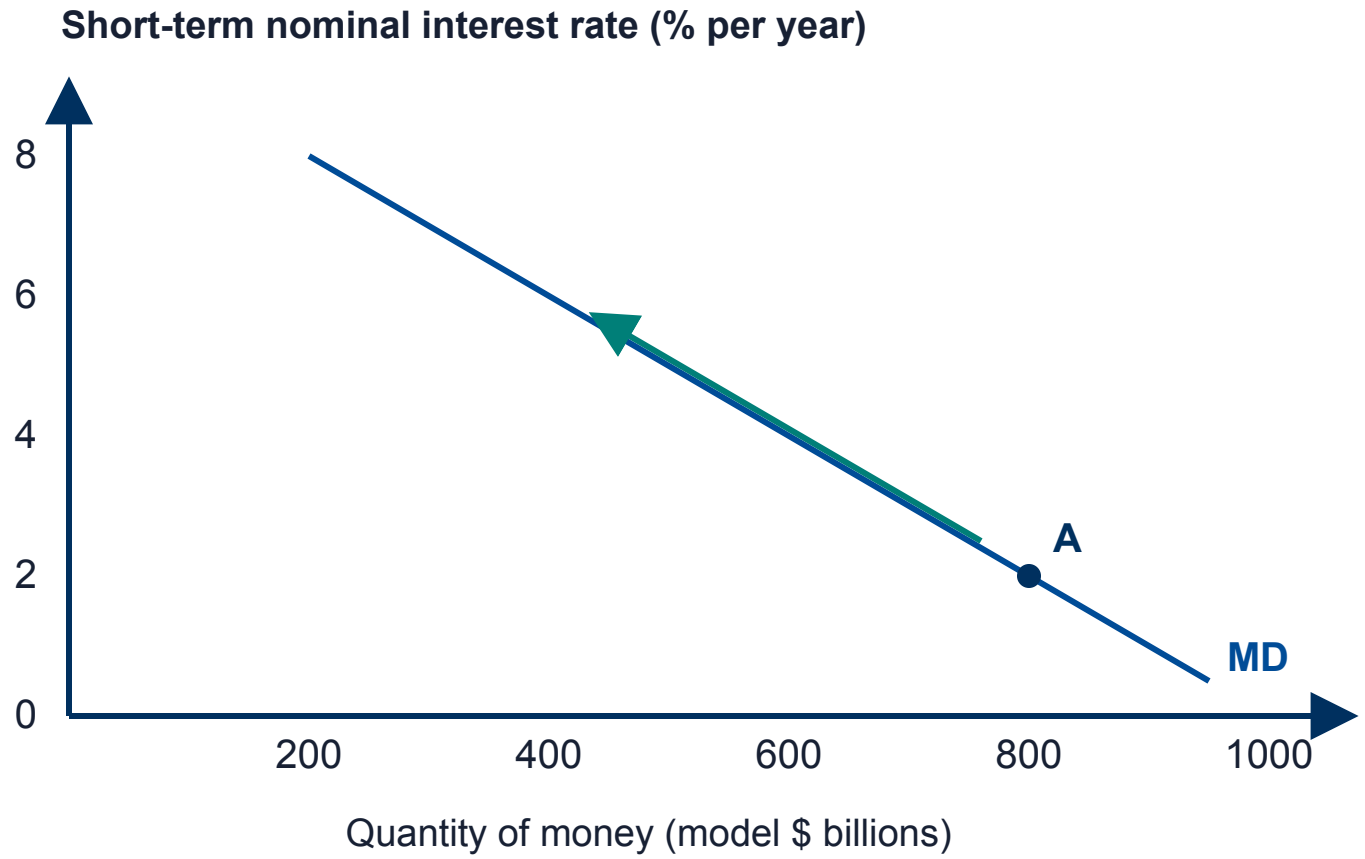
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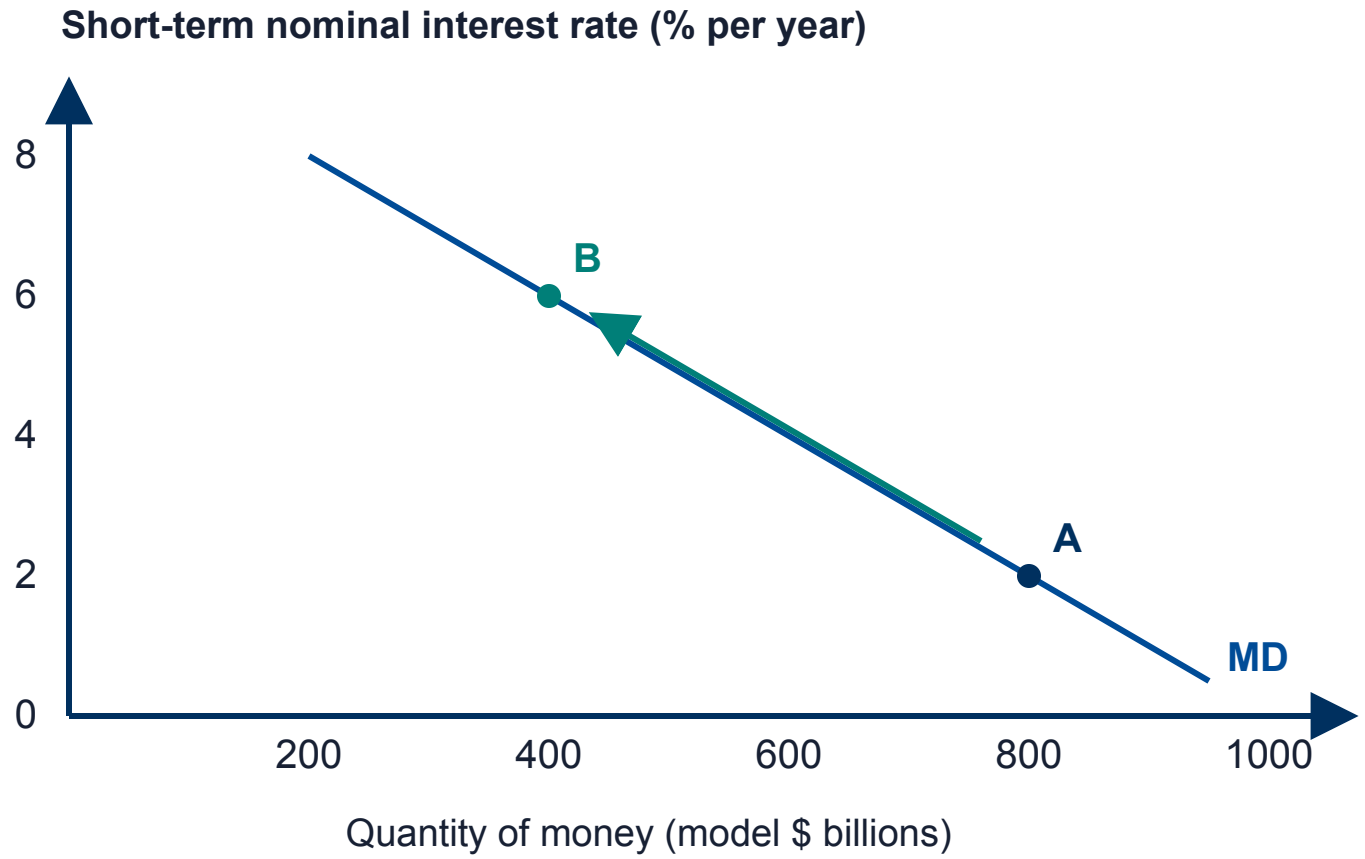
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3. Move upward along MD: less money is desired for liquidity at the higher rate.

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- Institutions that require larger cash buffers can raise desired balances → MD shifts right.

15.2 Money and Interest Rates

- Explain the federal funds rate and the Fed’s influence on short-term rates.
- Interpret money-market equilibrium and shifts in the supply of money.
- Explain why longer-term rates reflect expectations and need not move one-for-one with today’s policy rate.

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- The FOMC announces a target range; the market determines the effective rate.
- The Fed influences this overnight rate, which helps shape other short-term rates and broader financial conditions.

From a Fed Purchase to Money Supply

1. Recall the previous class: the Fed buys a Treasury security from a nonbank seller.

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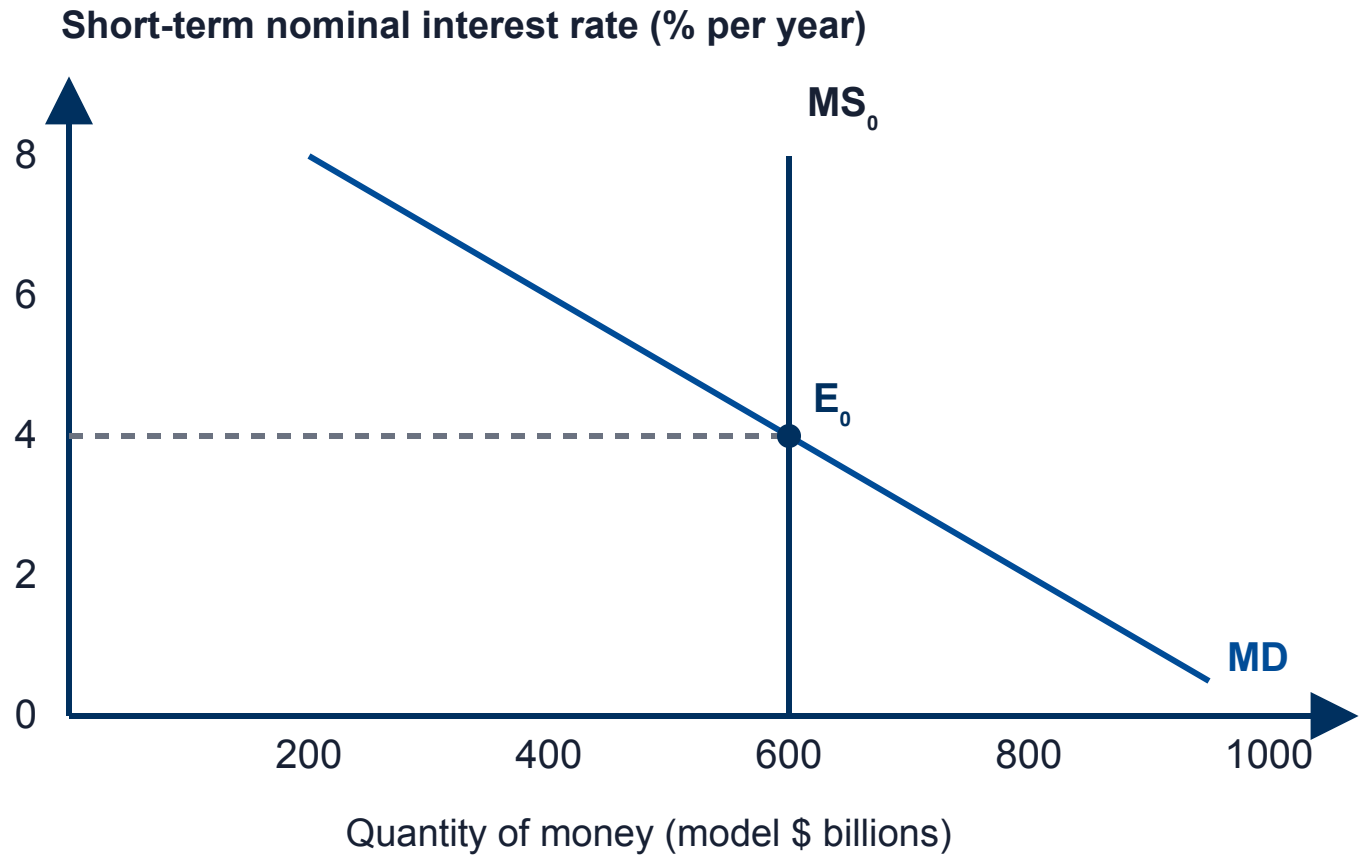
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4. The interest rate adjusts along the existing MD curve. Policy did not directly shift MD.

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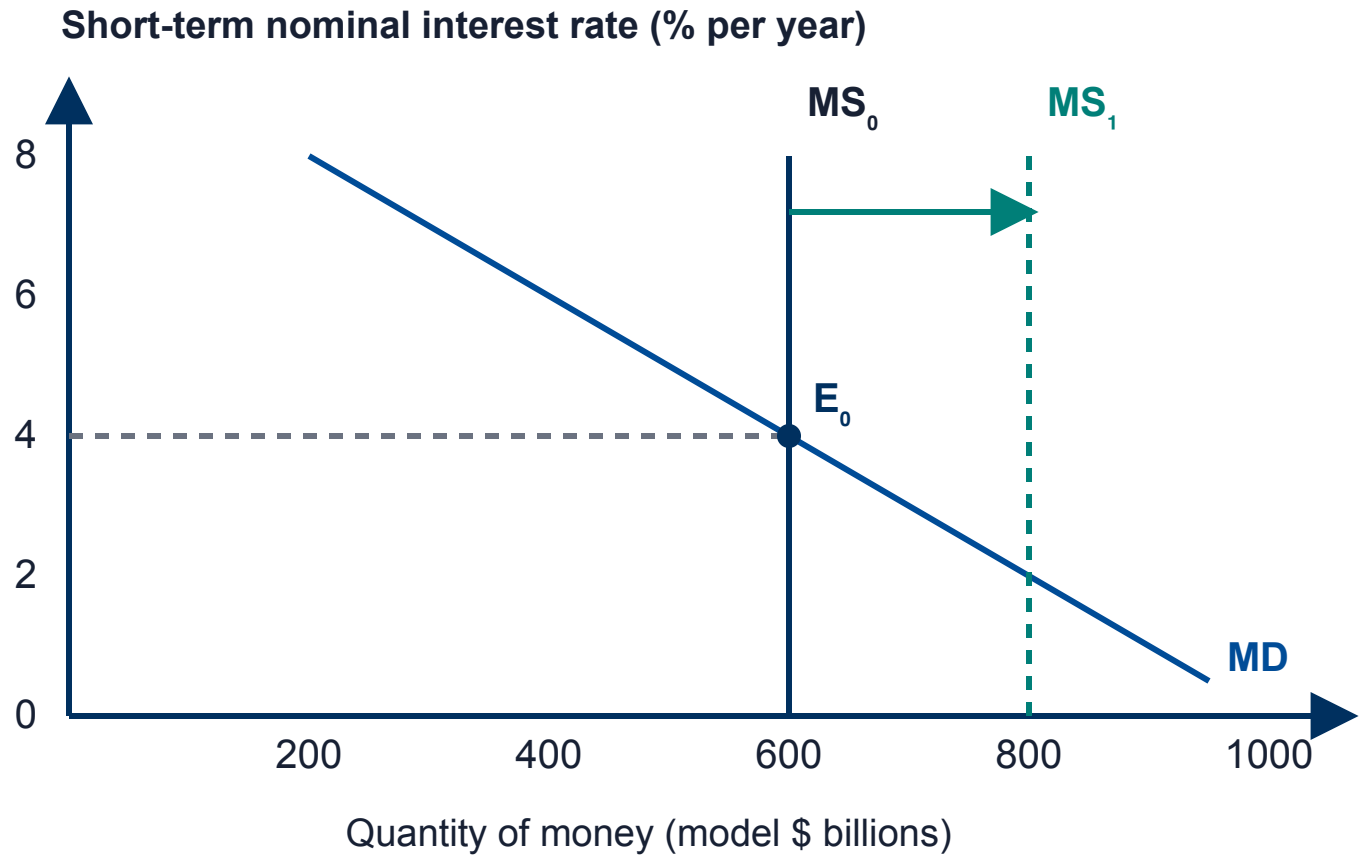
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1. Start at $MS_0 = 600$ and $i_0 = 4\%$.

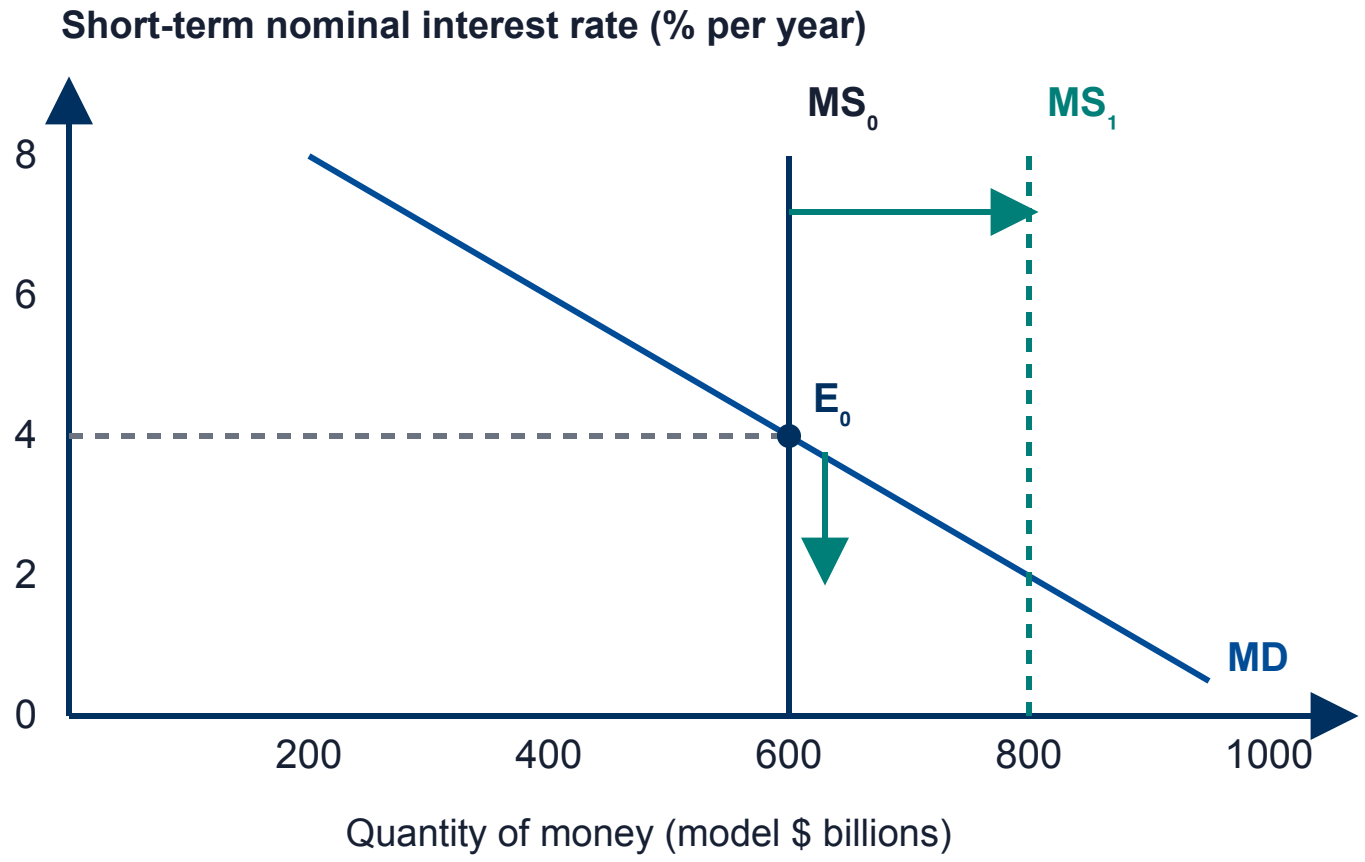
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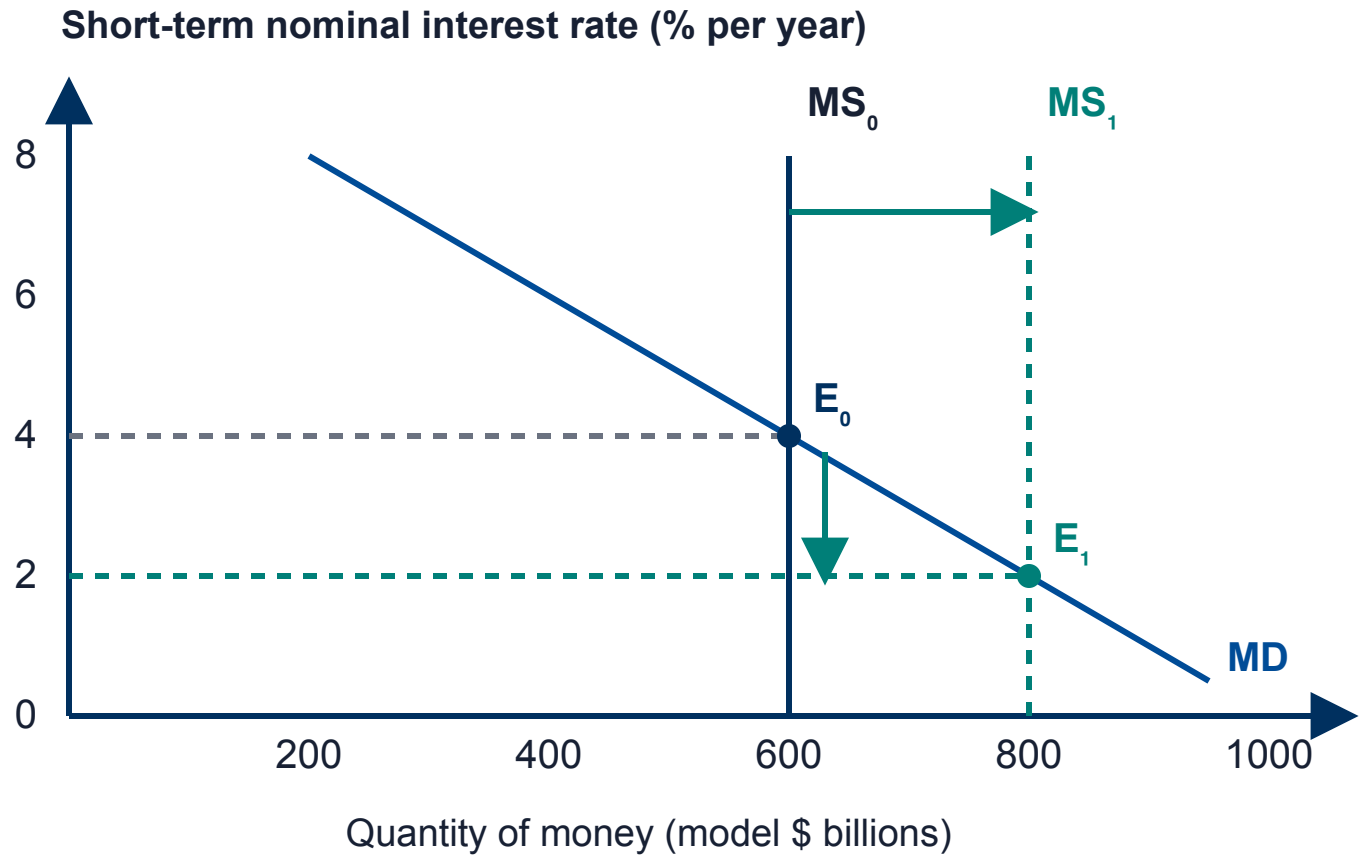
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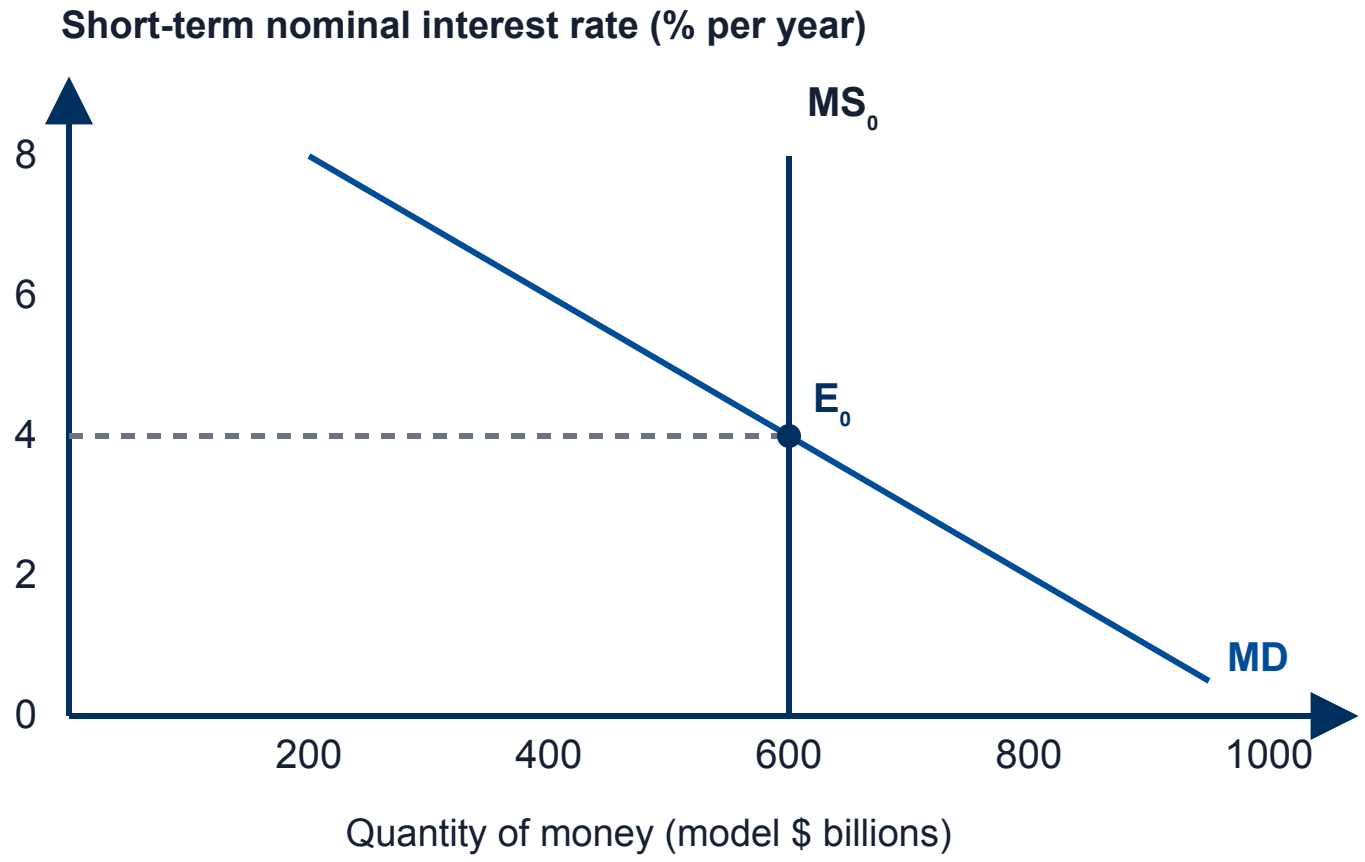
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4. Bond prices rise and yields fall. New equilibrium: $i_1 = 2\%$, $M_1 = 800$.

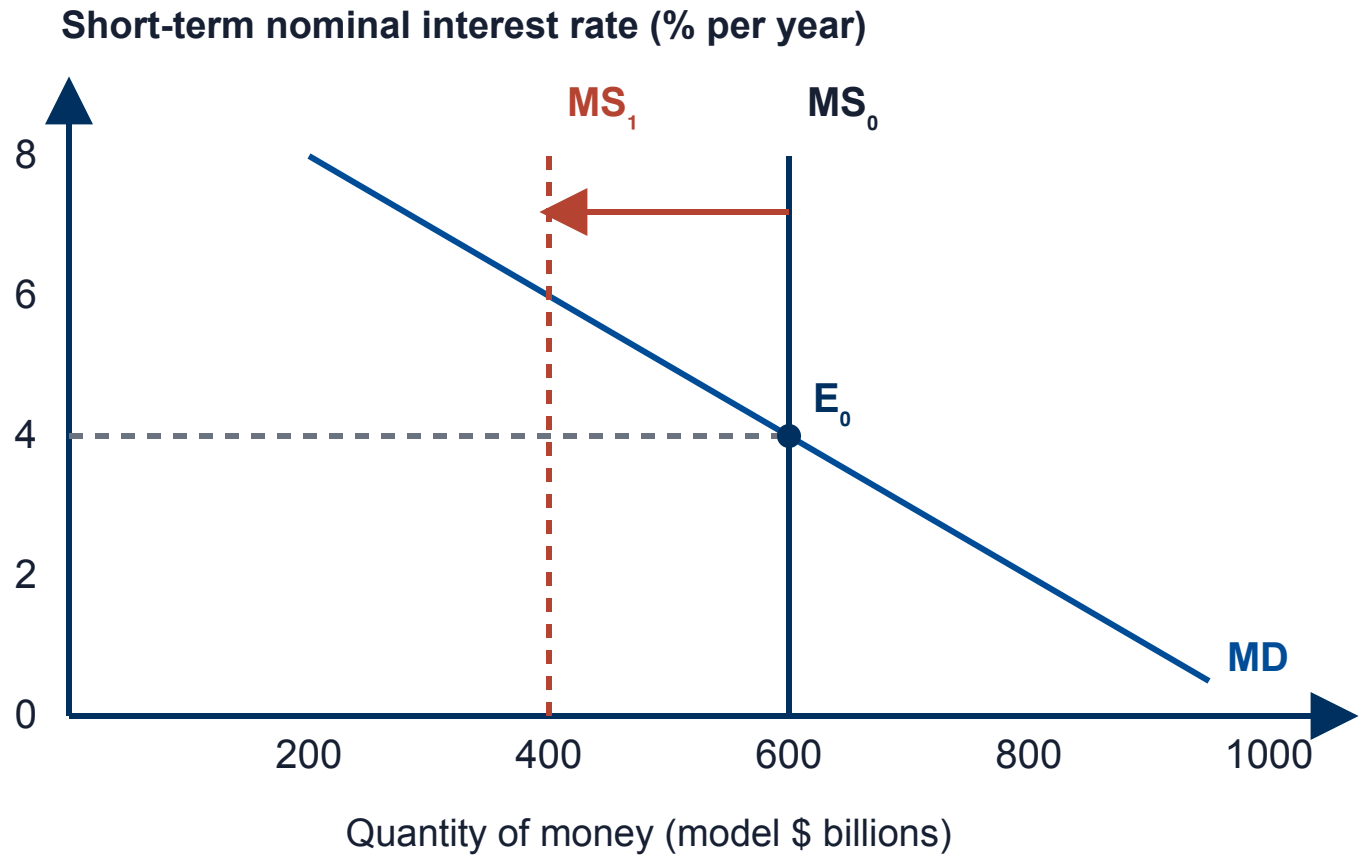
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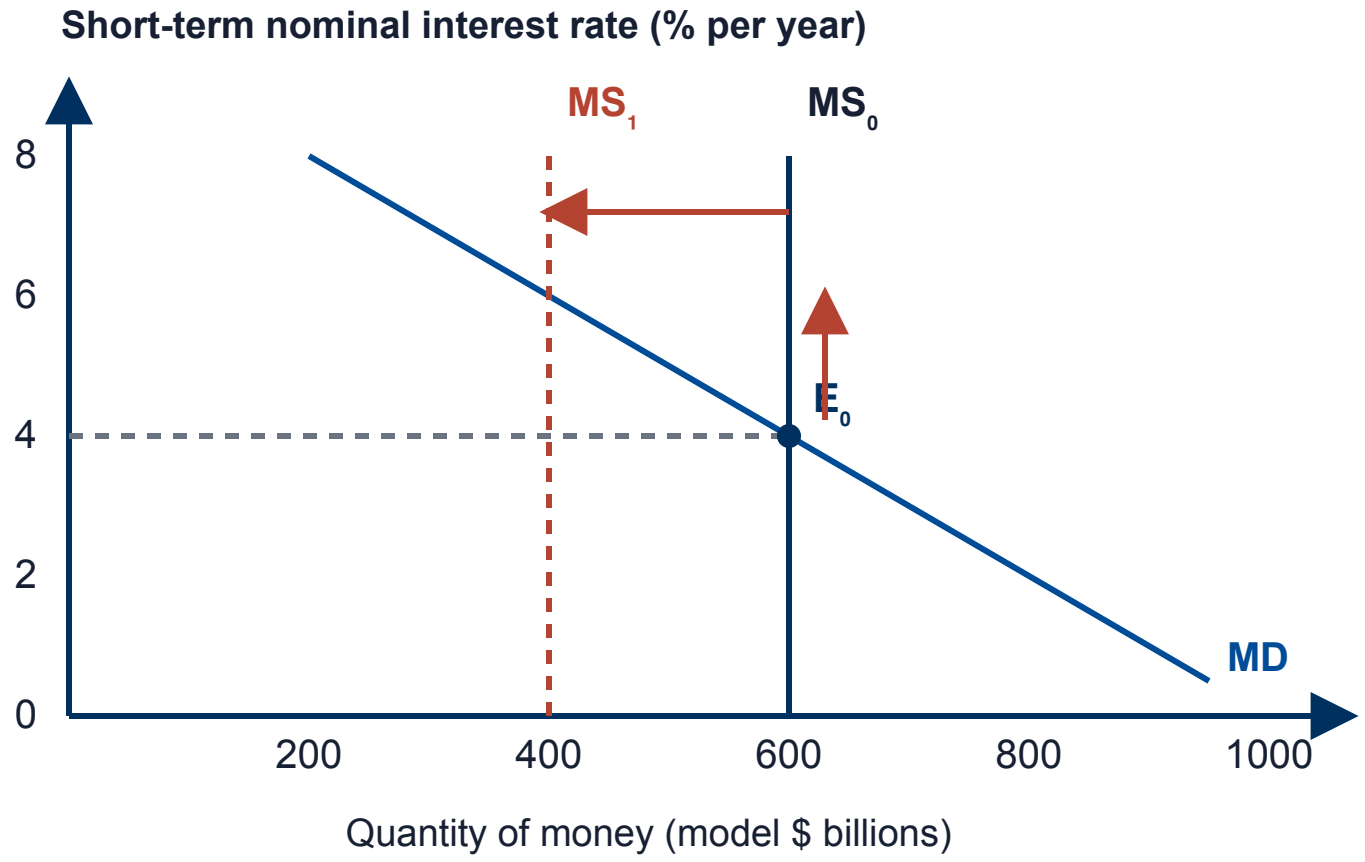
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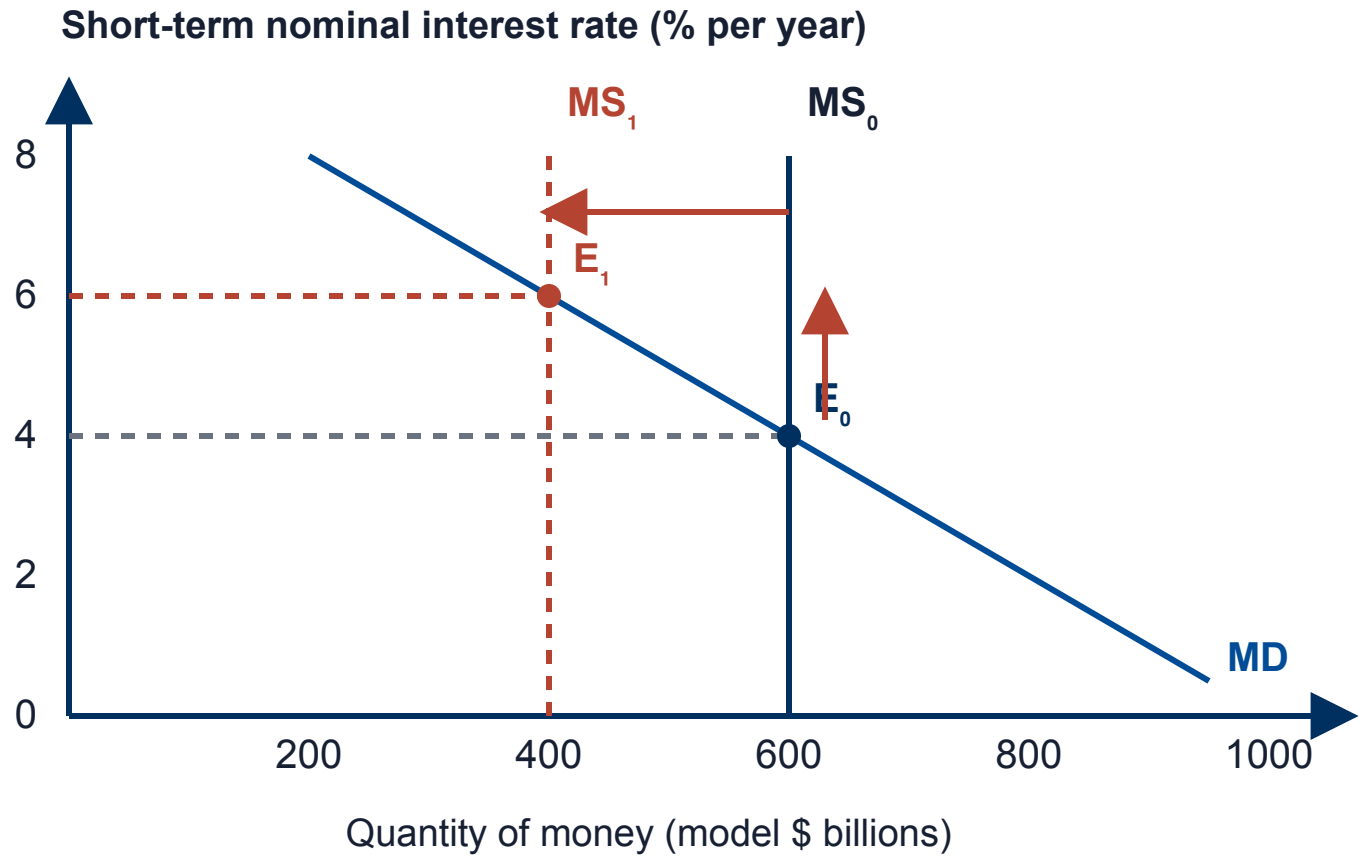
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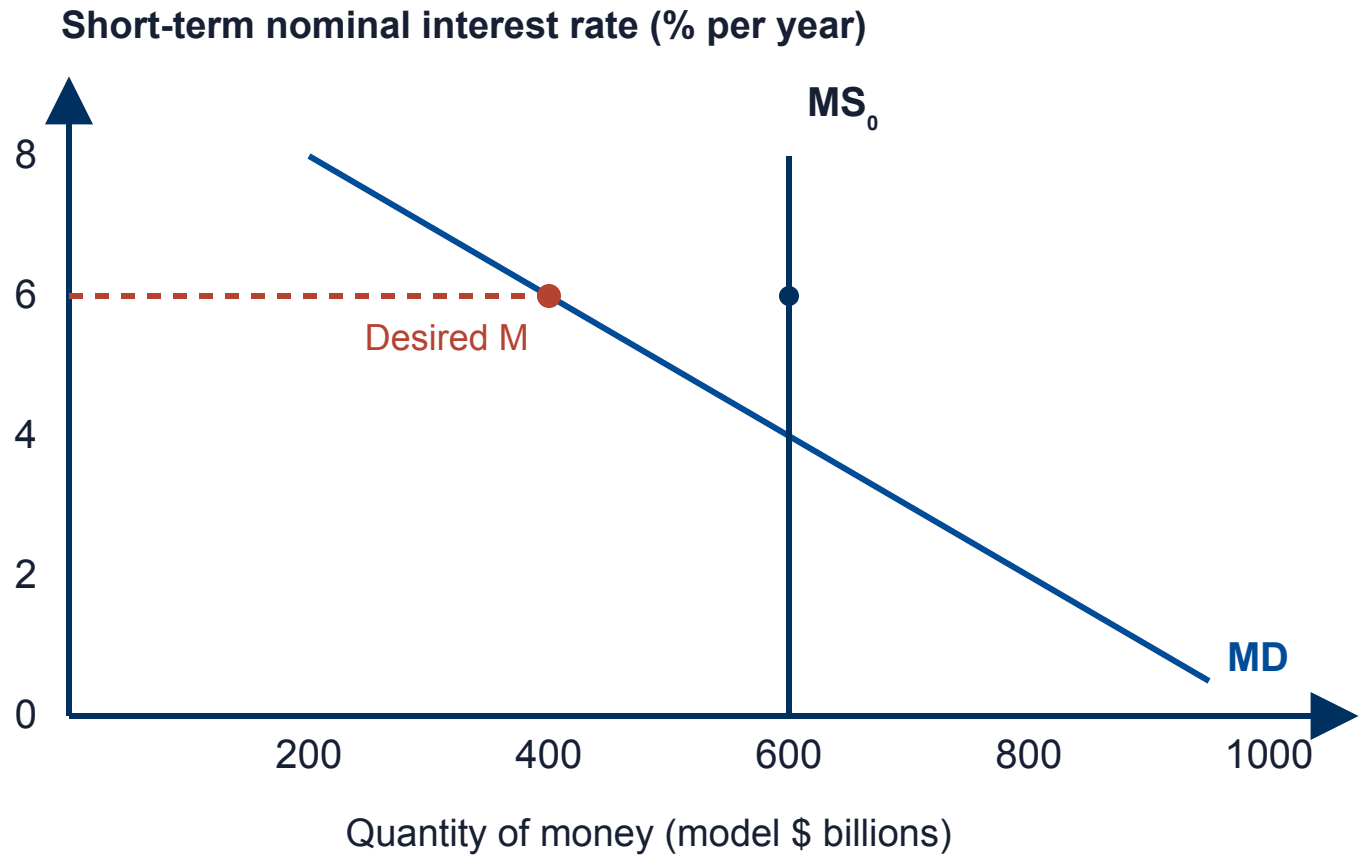
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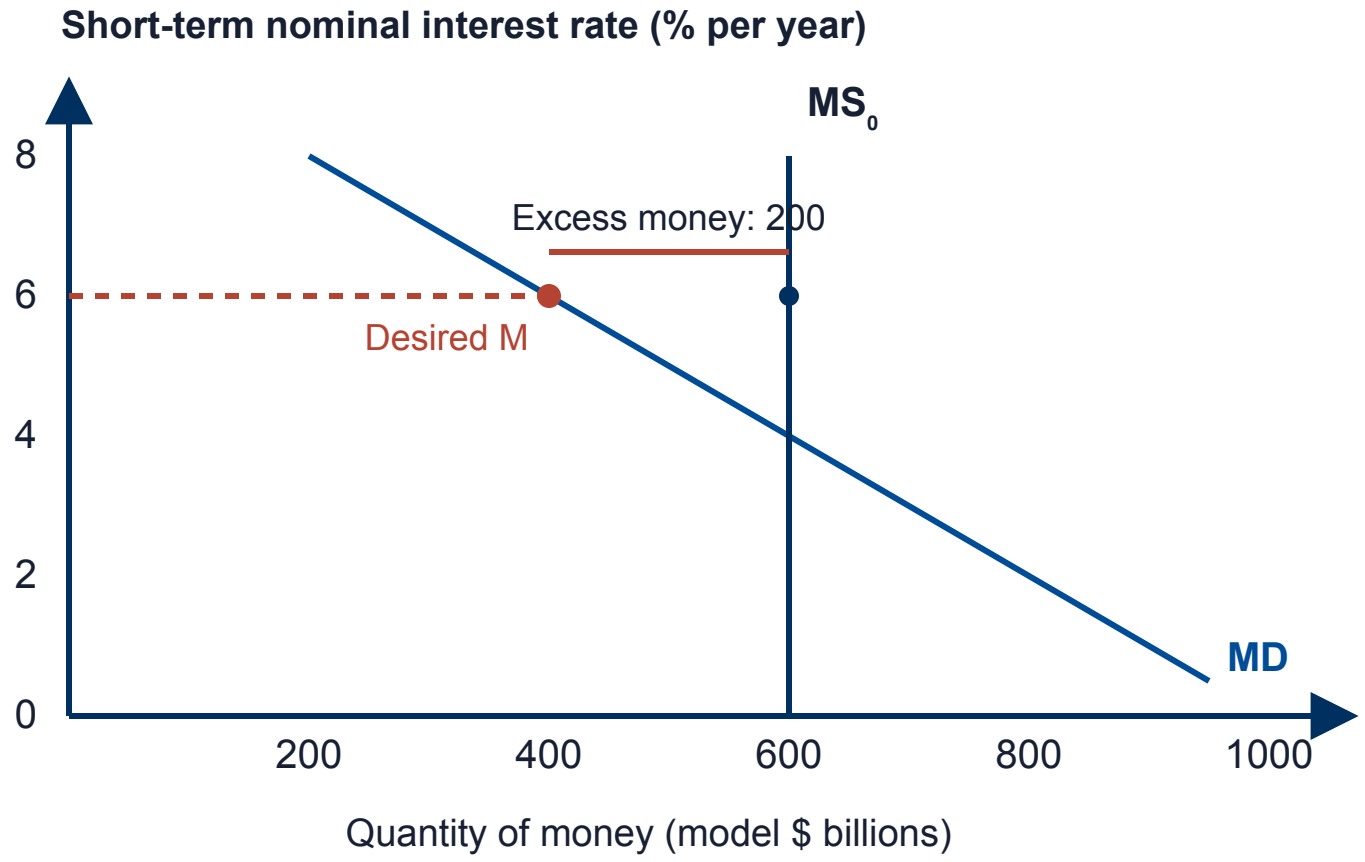
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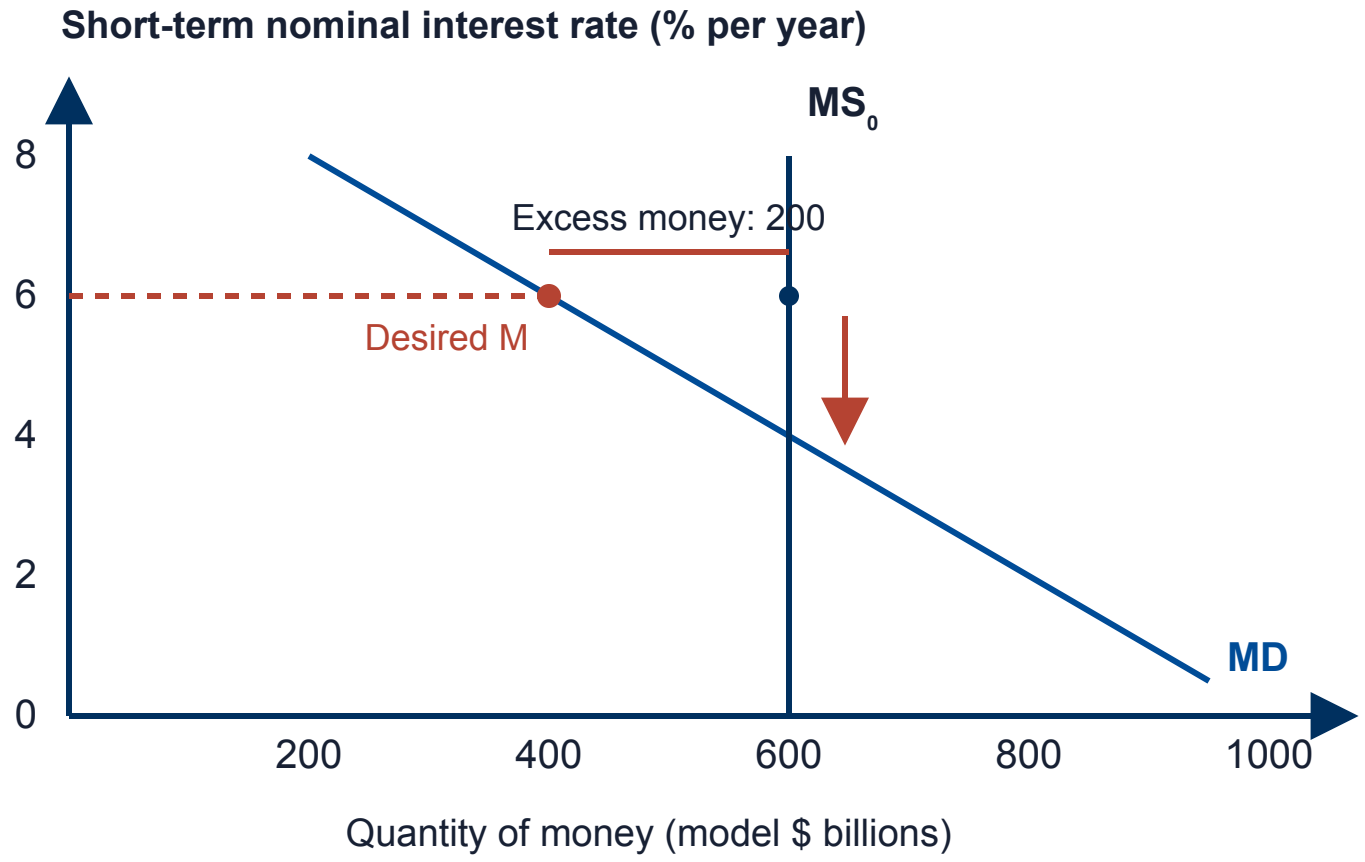
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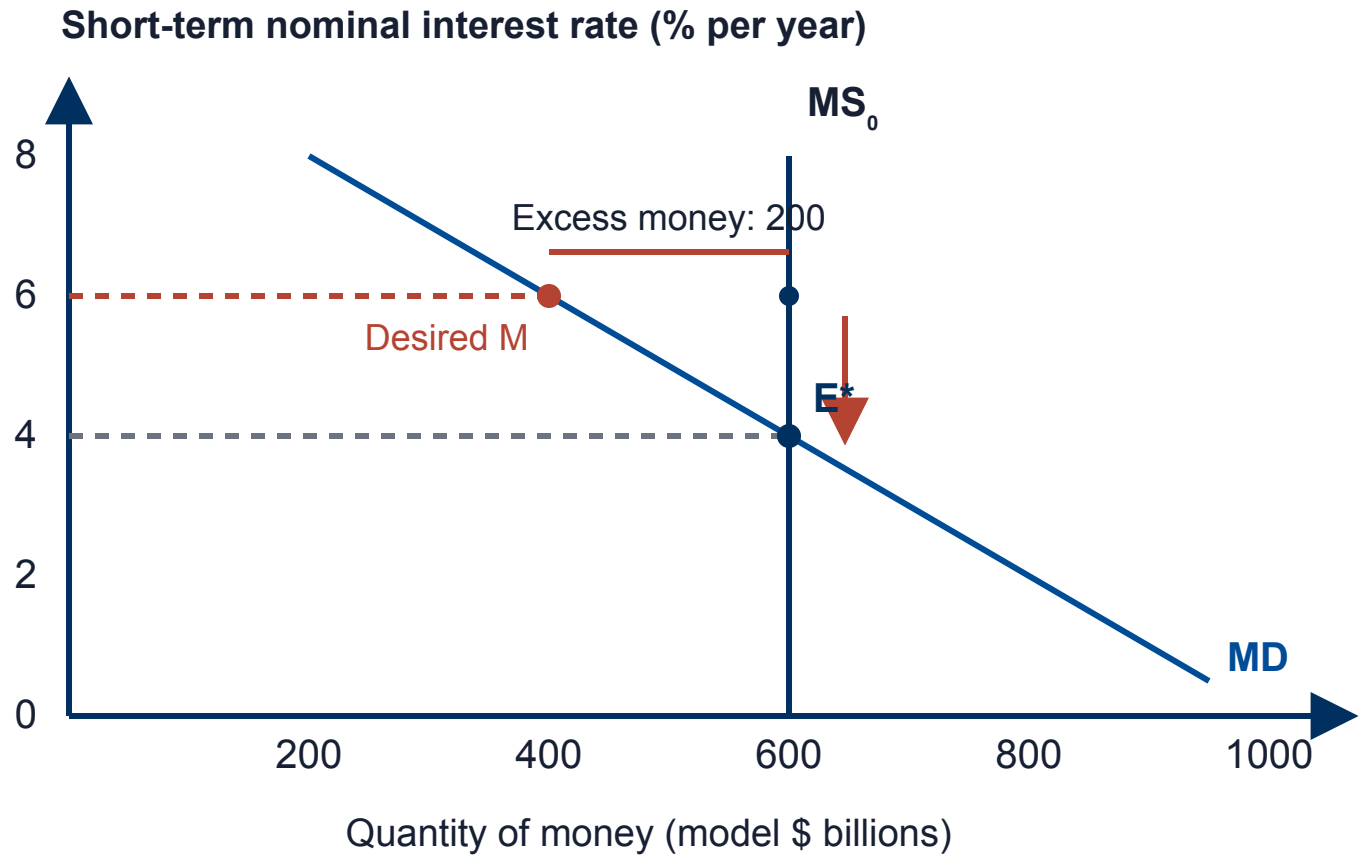
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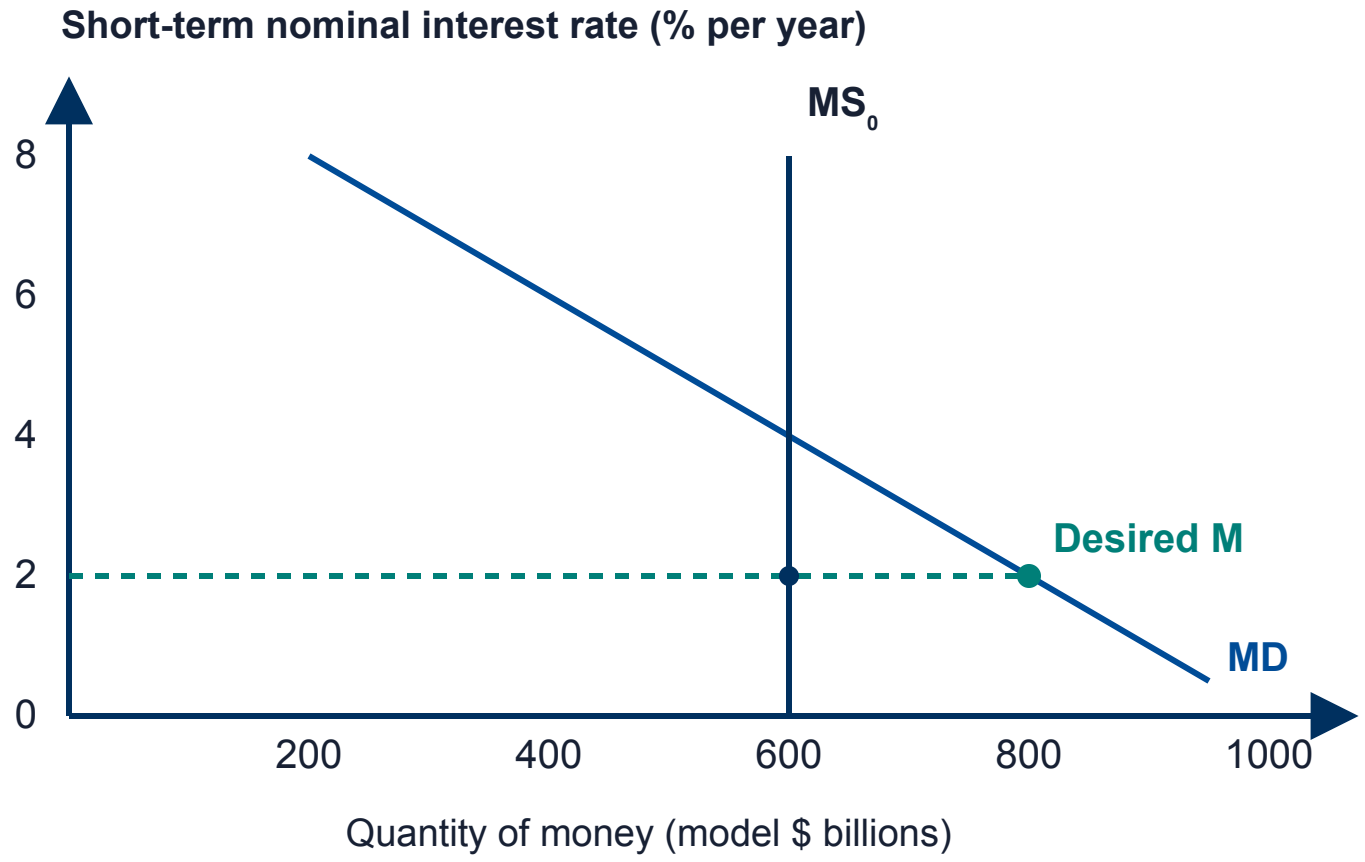
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4. At i = 4%, desired and available money both equal 600: the adjustment pressure ends.

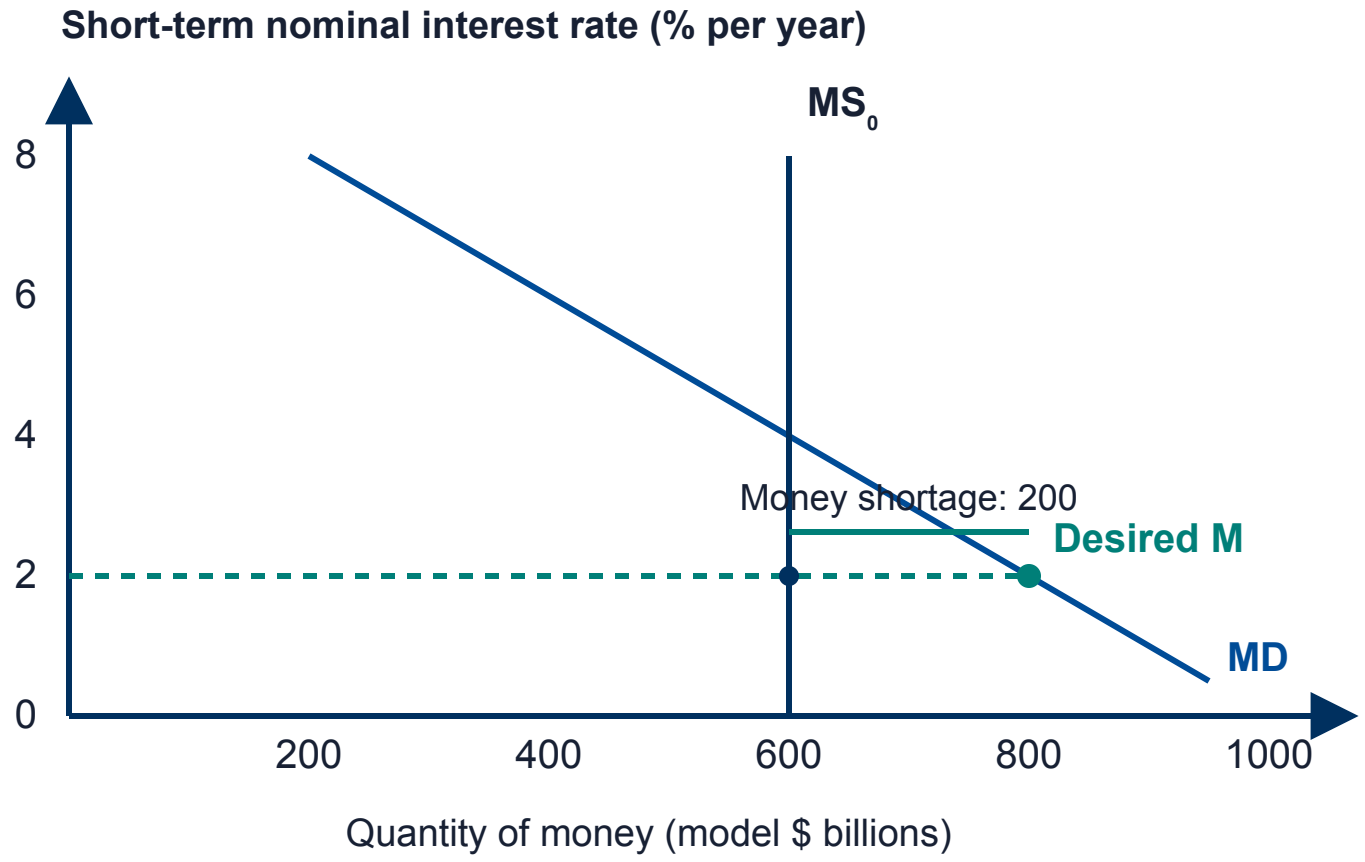
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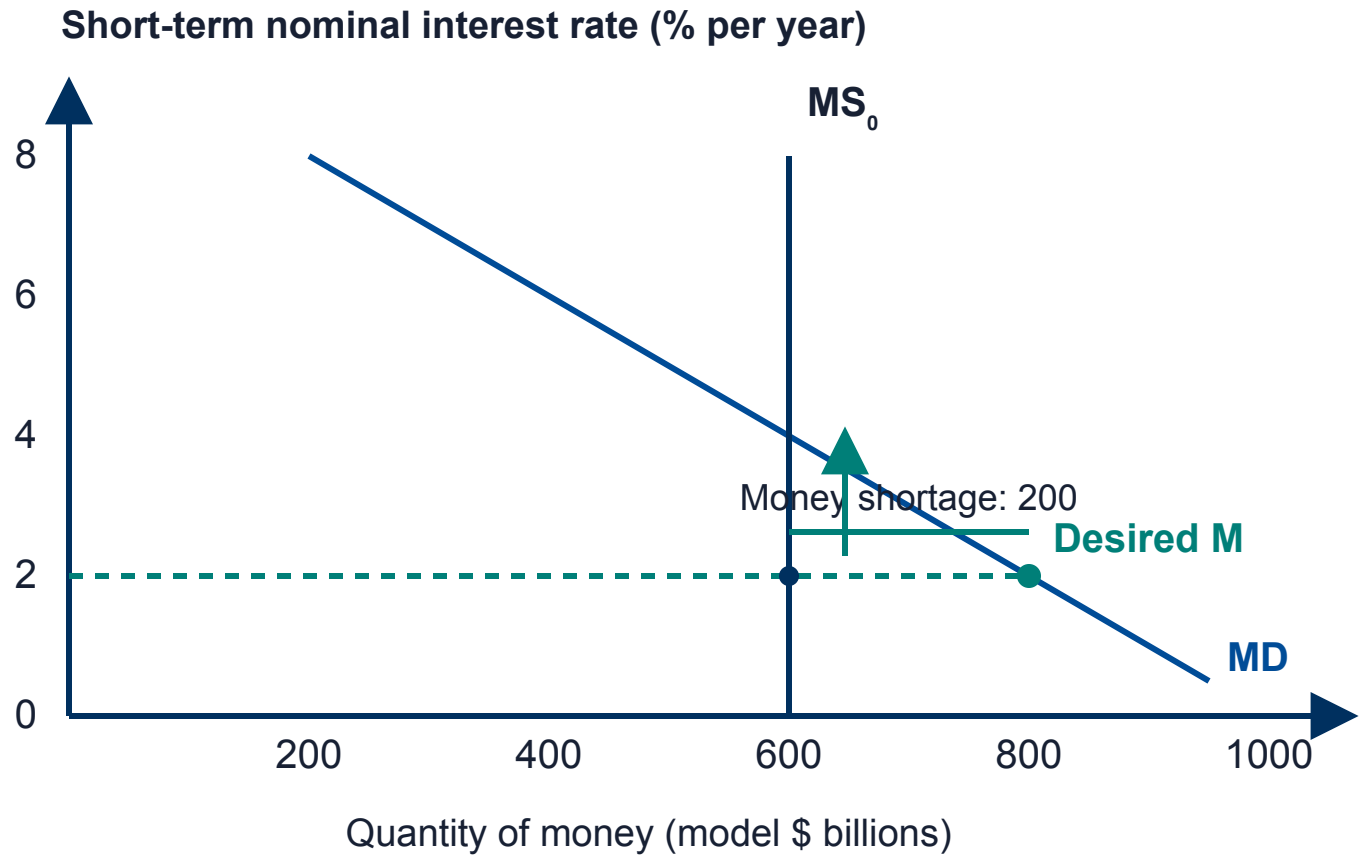
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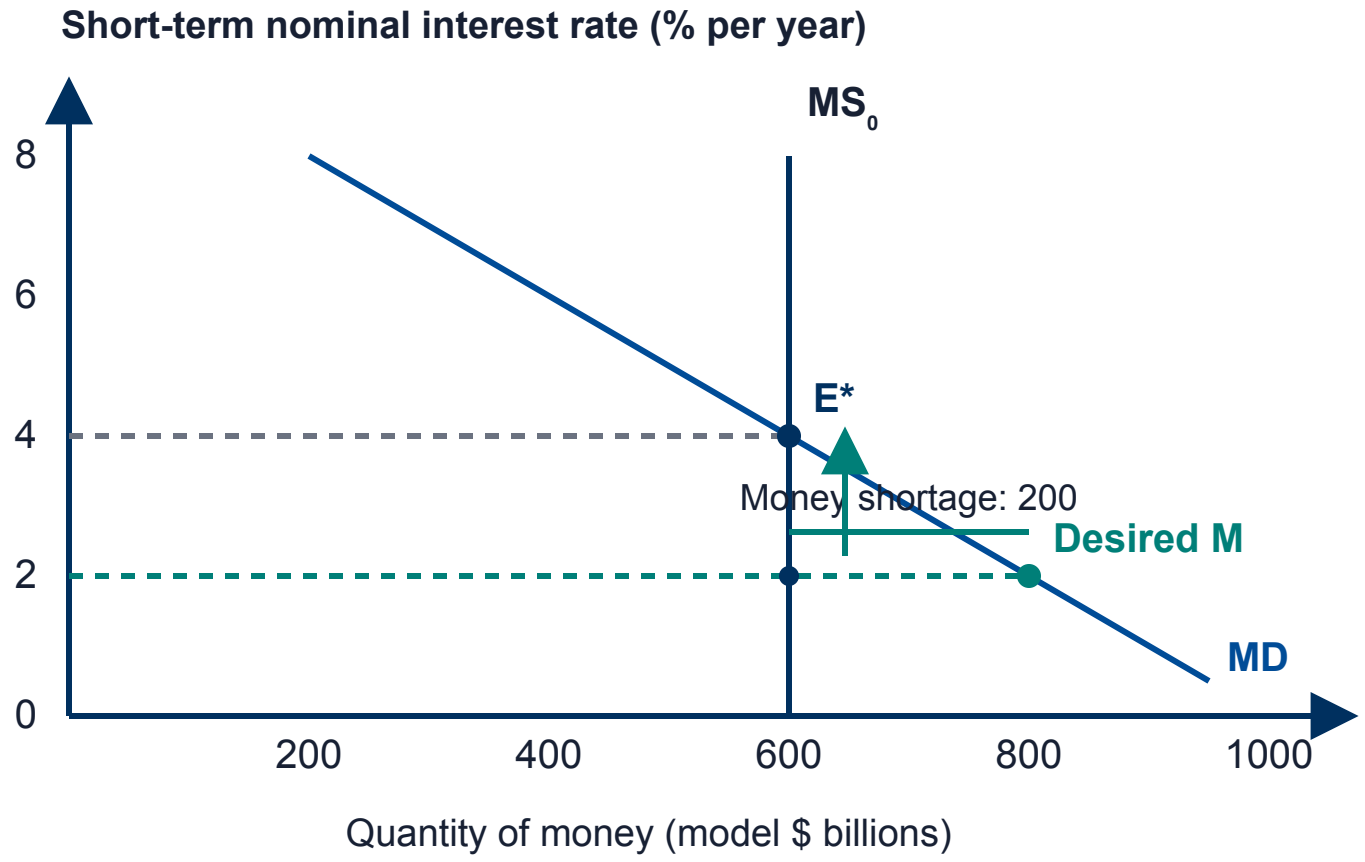
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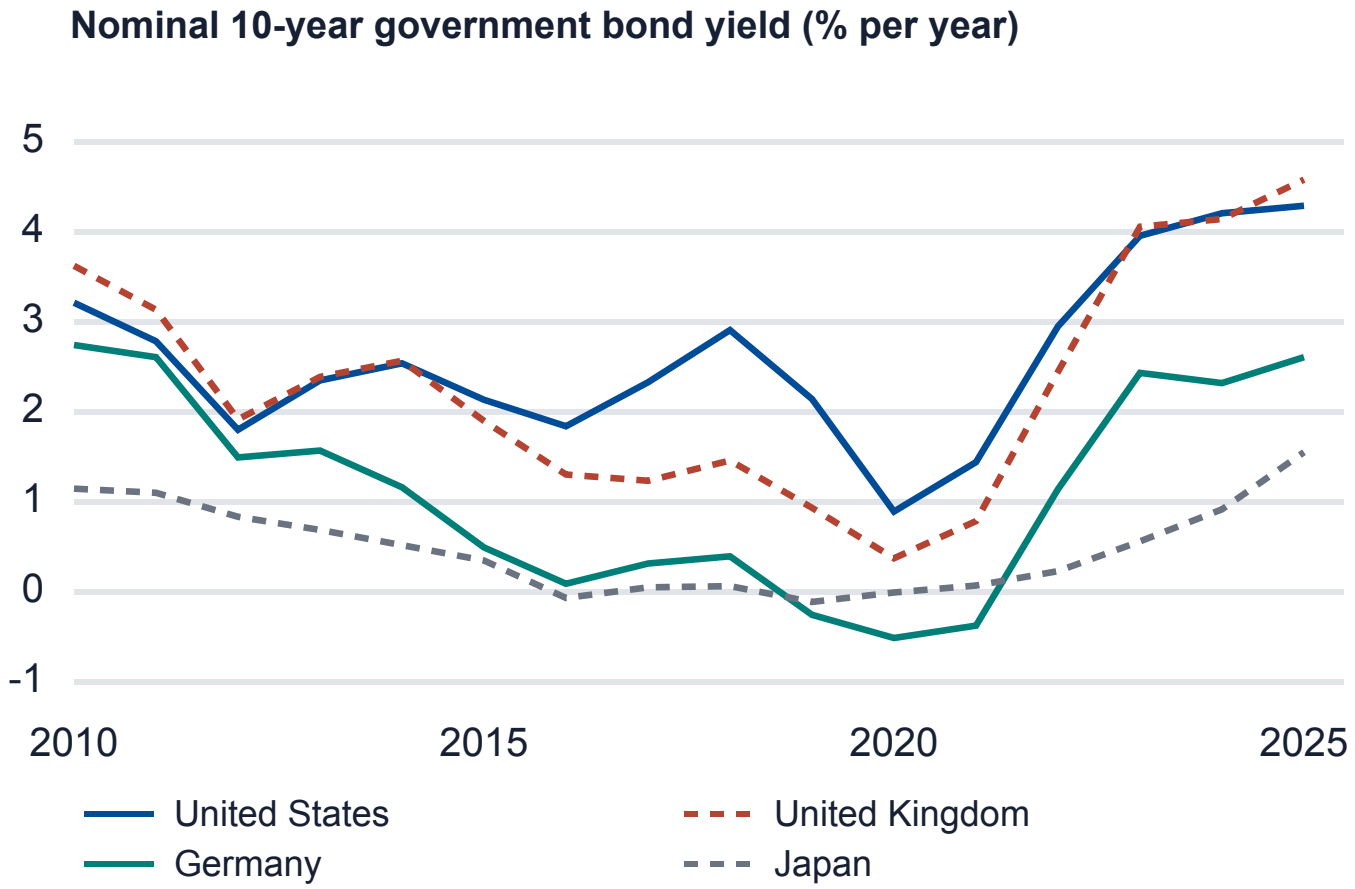
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4. At i = 4%, desired and available money both equal 600.

Money-Market Effects

Policy in the model	Money supply	Bond price	Short-term rate
Expansionary purchase	Increases	Rises	Falls
Contractionary sale	Decreases	Falls	Rises

Hold money demand, the price level, real GDP, and money’s own return fixed.

Long-Term Interest Rates across Countries

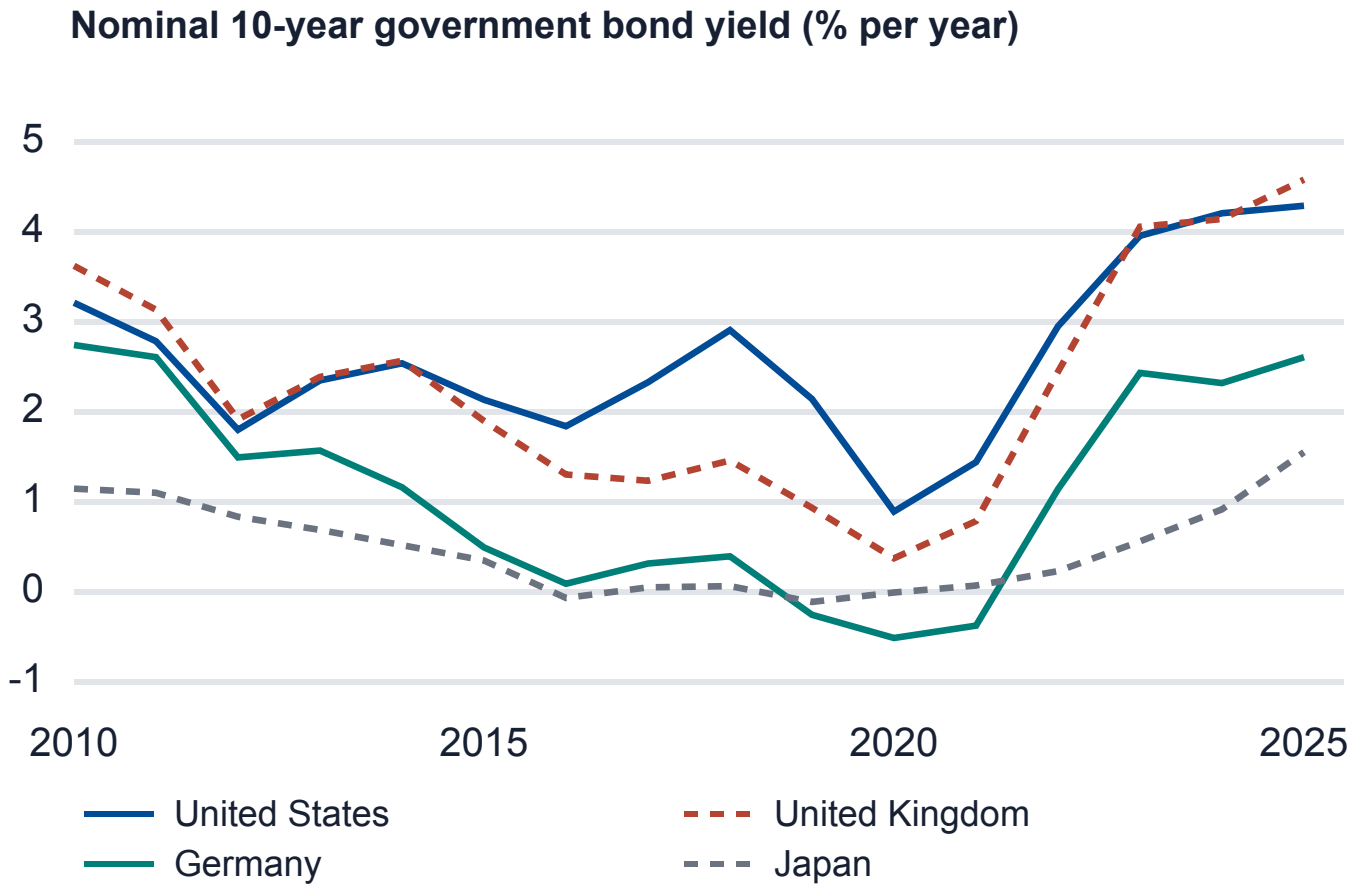


Annual averages, 2010–2025; own-currency yields. Original chart from OECD data.

1. Read first: annual-average yields on approximately 10-year government bonds, in each country's currency.

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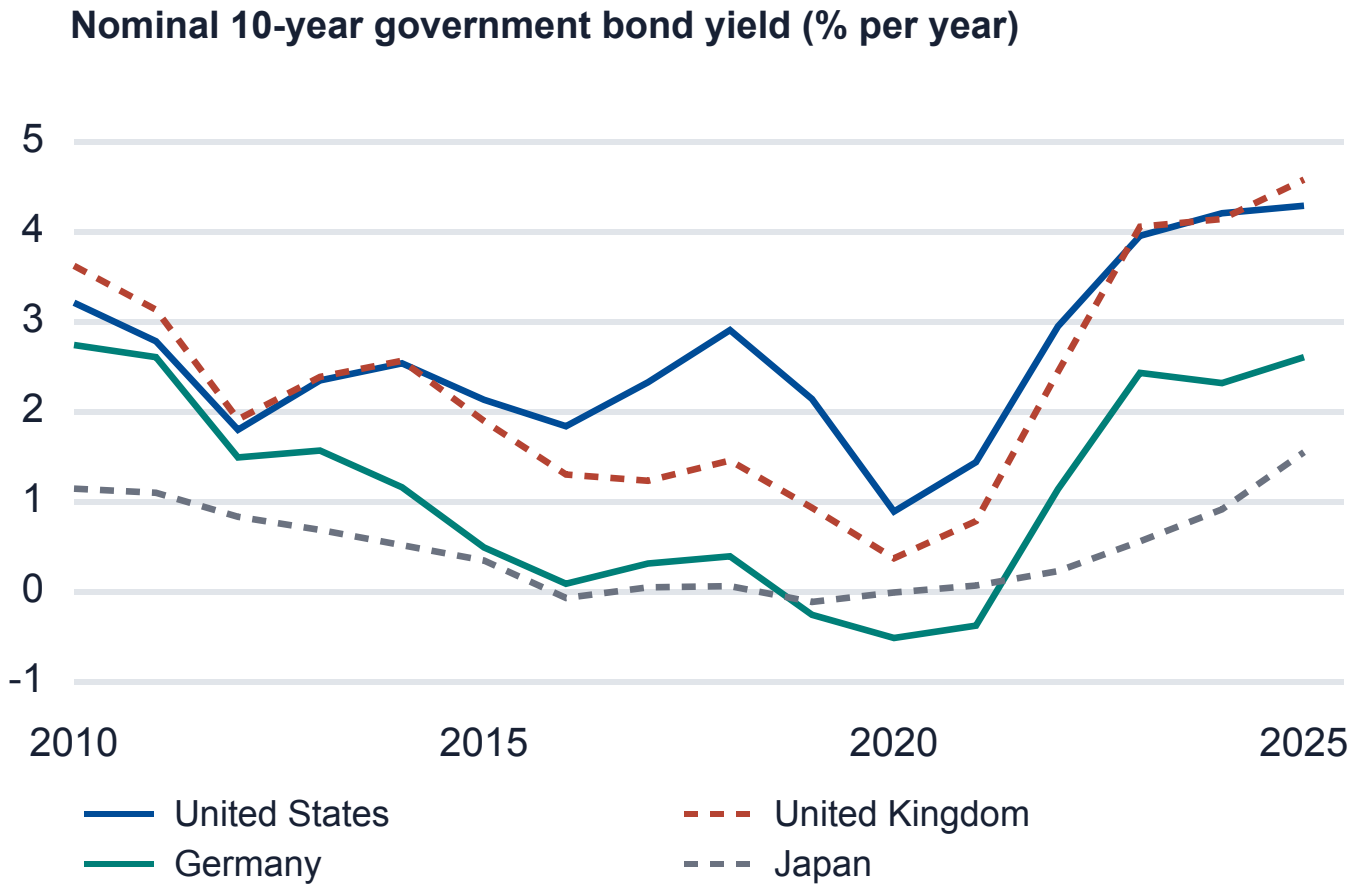


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1. Read first: annual-average yields on approximately 10-year government bonds, in each country's currency.
2. Compare Japan and the United States: their yields differ despite the same broad maturity.
3. The rates rise after 2021 in all four series, by different amounts. The chart alone does not identify the cause.

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One Year or Two Years?

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Two-year contract, annual compounding	5% fixed	5% fixed
Roll one-year investments: P5	4%	2% scenario
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- P6: rolling gives $\$600 \times 1.04 \times 1.08 = \$673.92$. The one-year strategy pays more.

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SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

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- A longer-term yield reflects the expected path of short-term rates and compensation for holding longer-term risk.
- A change in today’s policy rate may already be anticipated; a 10-year yield need not change by the same amount.

SEC Investor.gov: bonds · Federal Reserve Board: policy goals and transmission

Why Maturity Matters

- A longer-term yield reflects the expected path of short-term rates and compensation for holding longer-term risk.
- A change in today's policy rate may already be anticipated; a 10-year yield need not change by the same amount.
- Longer maturity is different from the macroeconomic long run. A two-year investment is longer than a one-year investment; the data chart uses a 10-year benchmark.

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15.3 Monetary Policy and Aggregate Demand

- Explain how monetary policy affects consumption, investment, and aggregate demand.
- Interpret policy-induced AD shifts while holding short-run aggregate supply fixed.
- Demonstrate expansionary and contractionary effects on output and the price level.

Fiscal and Monetary Policy

Policy choice	Main route	When especially useful
Monetary policy	Changes financial conditions; affects C and I	Broad demand management with room to adjust rates
Fiscal policy	Government purchases, taxes, and transfers	Direct purchases or support targeted to affected groups
A coordinated response	Combines instruments	Severe downturns, especially near the effective lower bound

- Automatic stabilizers can act without a new law; discretionary fiscal measures may take time to enact.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

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- The choice also depends on inflation, implementation delays, debt capacity, and the type of shock. A supply loss is not repaired simply by adding demand.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

From Interest Rates to Aggregate Demand

1. A lower policy rate can ease broader borrowing and financial conditions.

Transmission is conditional: expected inflation, credit access, risk, and confidence also matter.

Federal Reserve Board: policy goals and transmission

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Federal Reserve Board: policy goals and transmission

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Federal Reserve Board: policy goals and transmission

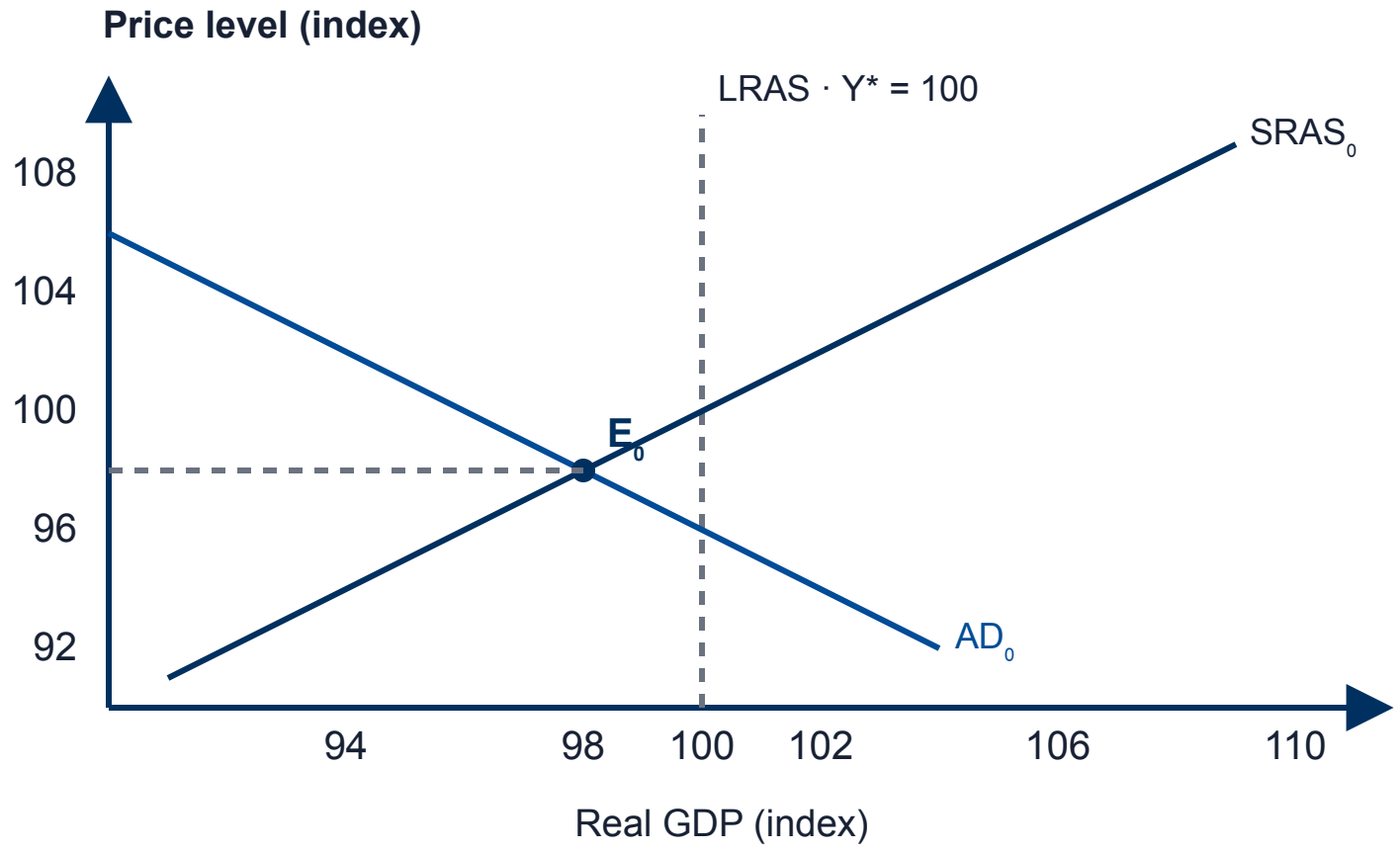
From Interest Rates to Aggregate Demand

1. A lower policy rate can ease broader borrowing and financial conditions.
2. Lower relevant real borrowing costs can encourage consumption and investment.
3. A $10,000 project paying $10,500 next year loses money with a $10,600 repayment but gains with a $10,400 repayment.
4. At a given price level, higher C and I raise planned expenditure: AD shifts right.

Transmission is conditional: expected inflation, credit access, risk, and confidence also matter.

Federal Reserve Board: policy goals and transmission

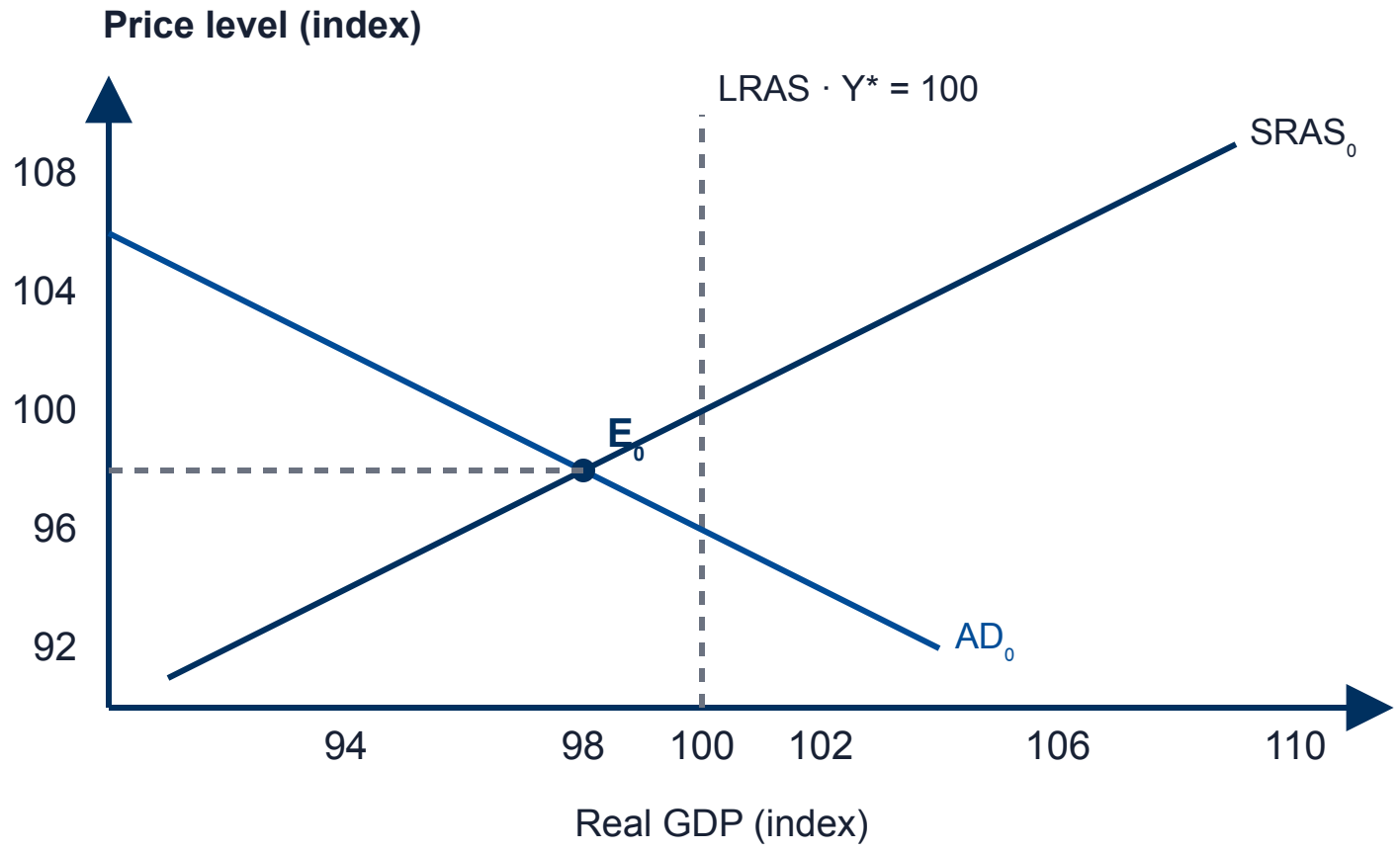
Expansionary Policy and Aggregate Demand



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Begin below potential output: $Y_0 = 98$, while $Y^* = 100$.

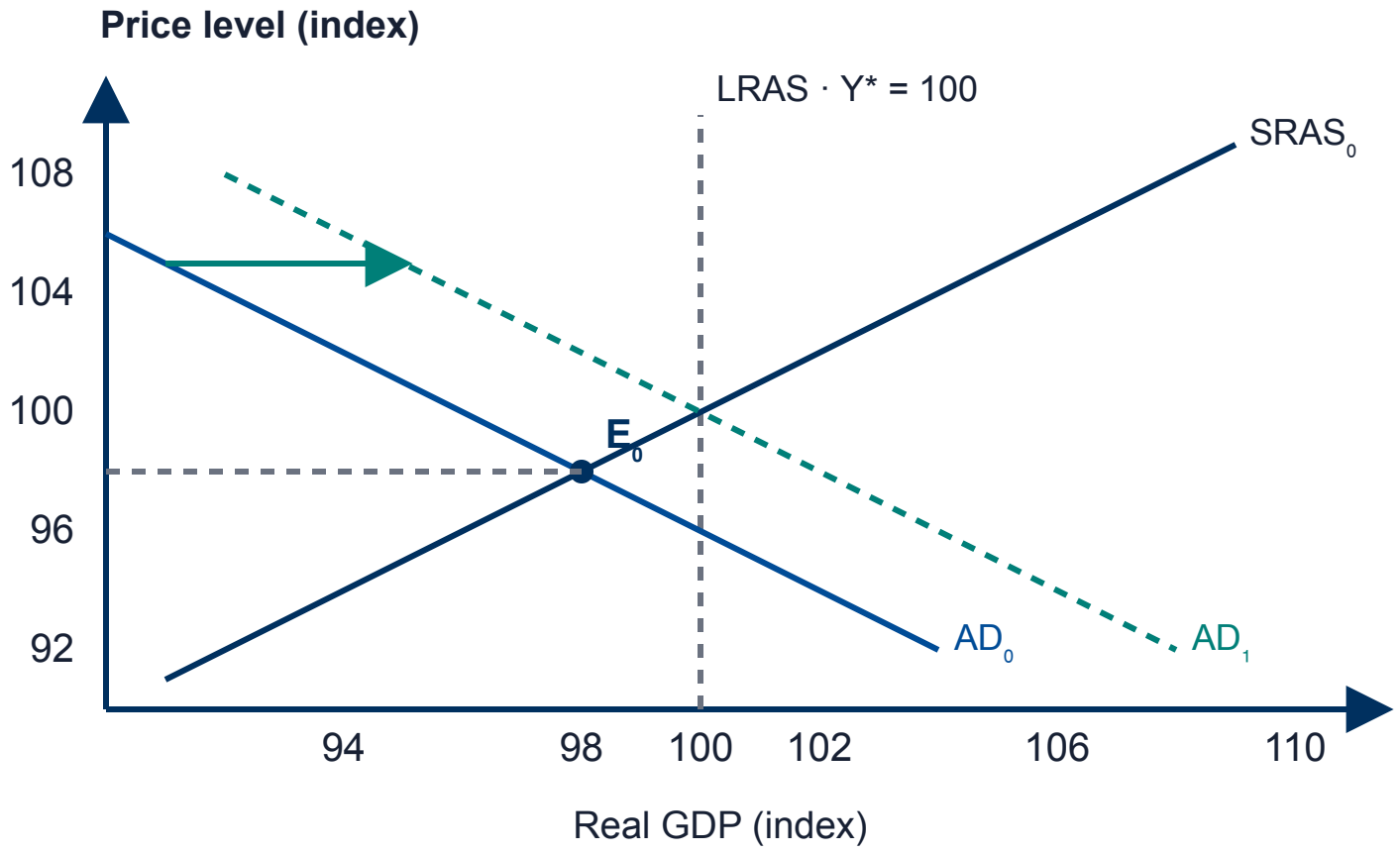
Expansionary Policy and Aggregate Demand



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1. Begin below potential output: $Y_0 = 98$, while $Y^* = 100$.
2. Easier financial conditions encourage interest-sensitive C and I.

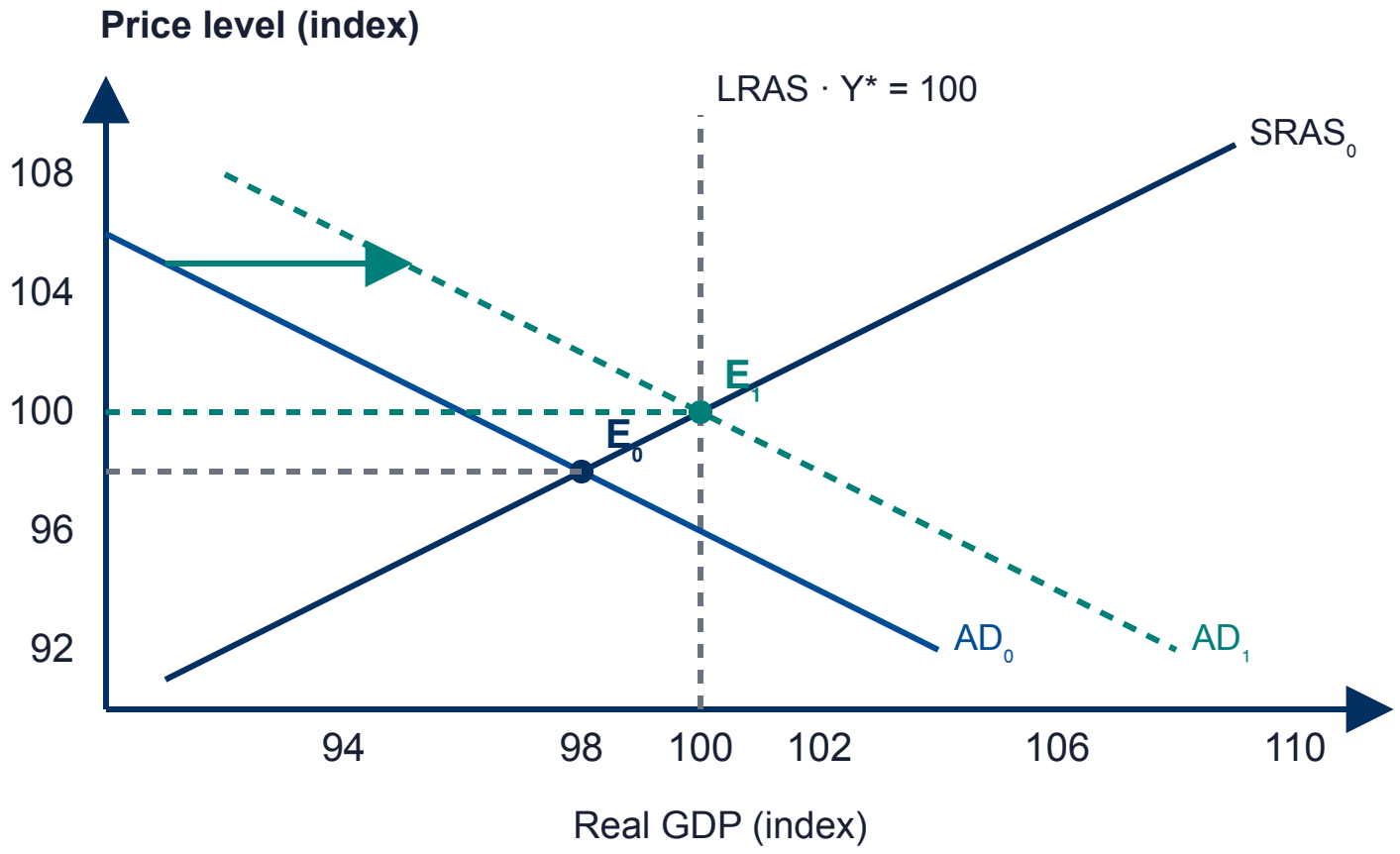
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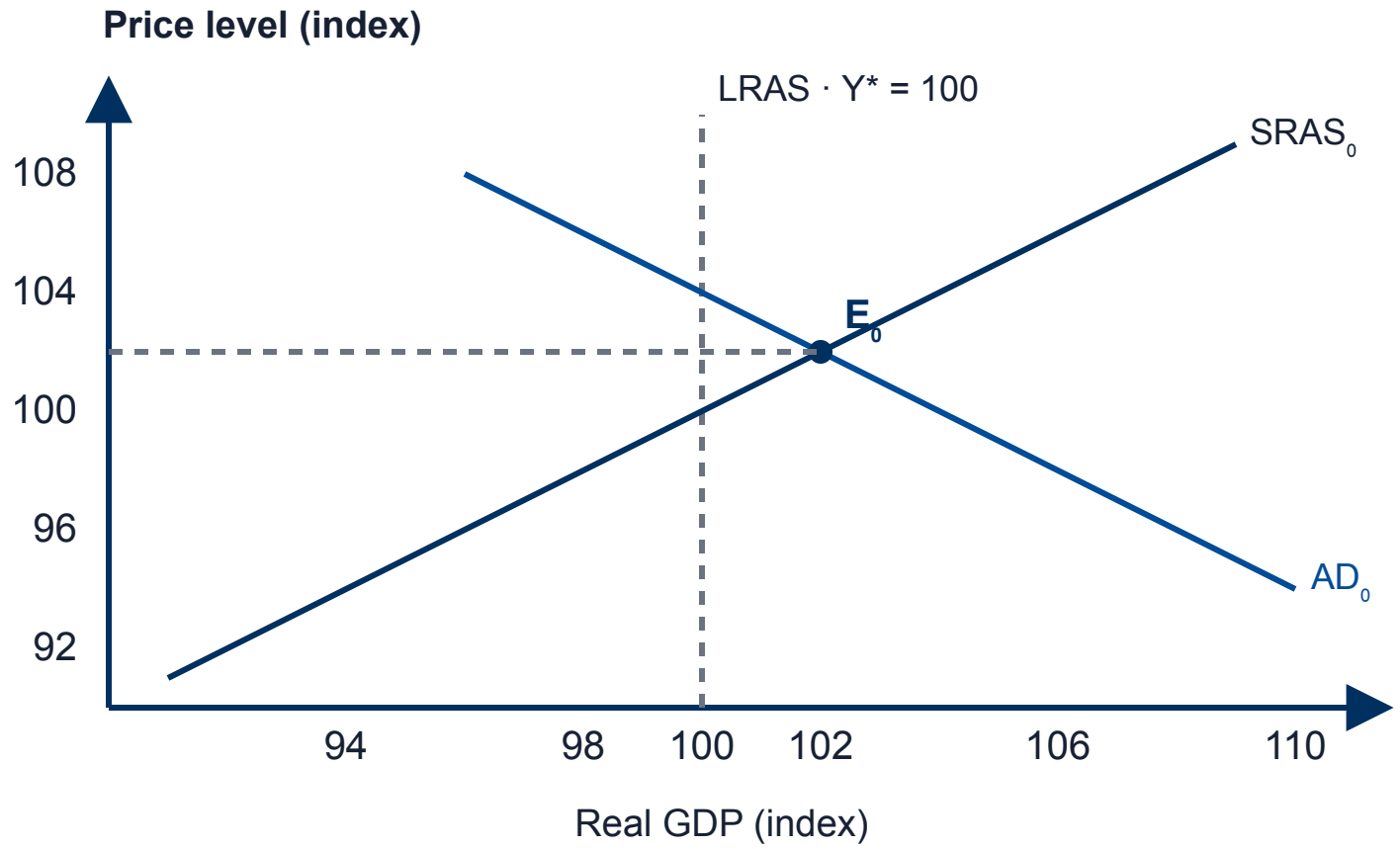
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4. In this short-run model, output and the price level rise to 100.

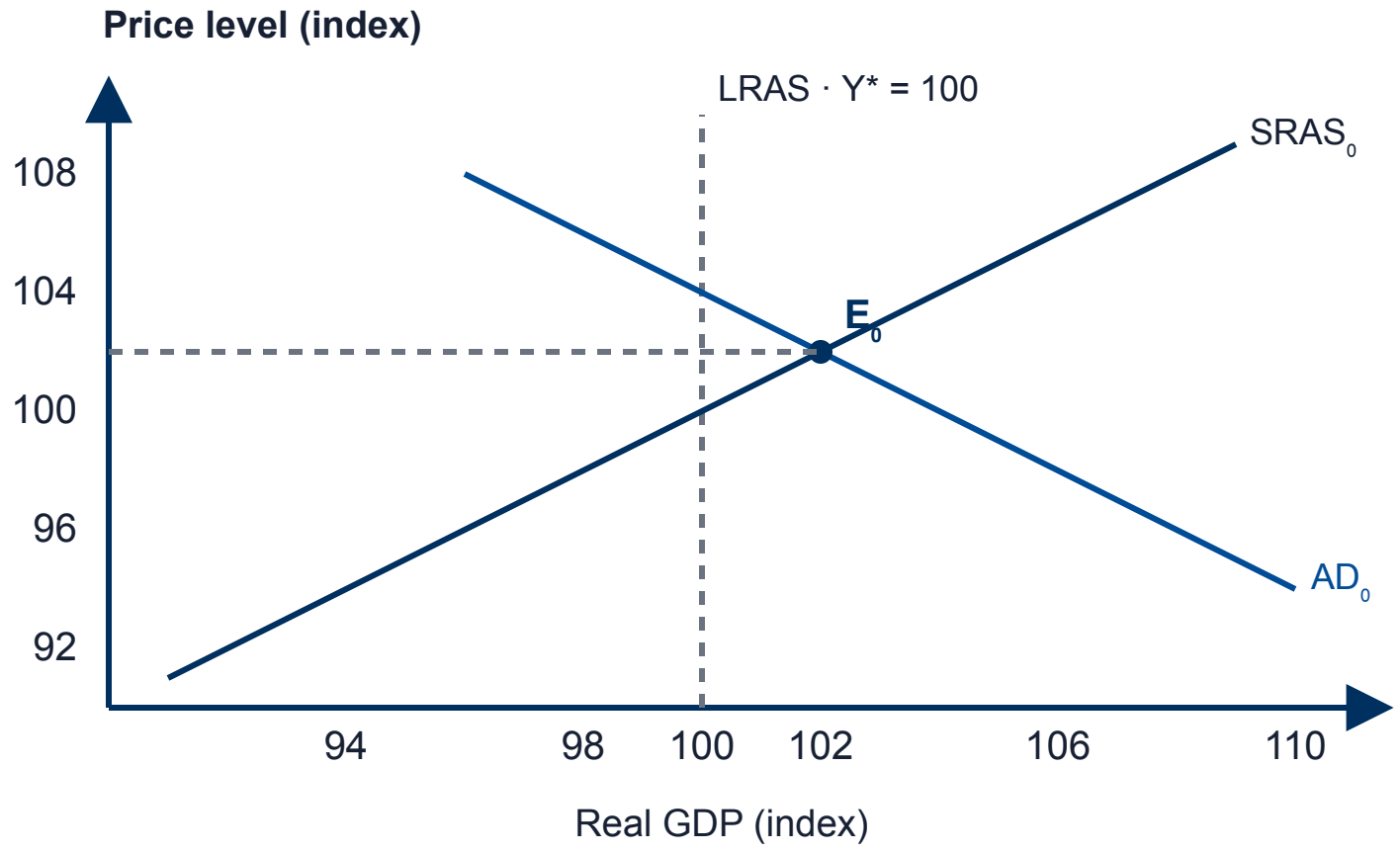
Contractionary Policy and Aggregate Demand



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.

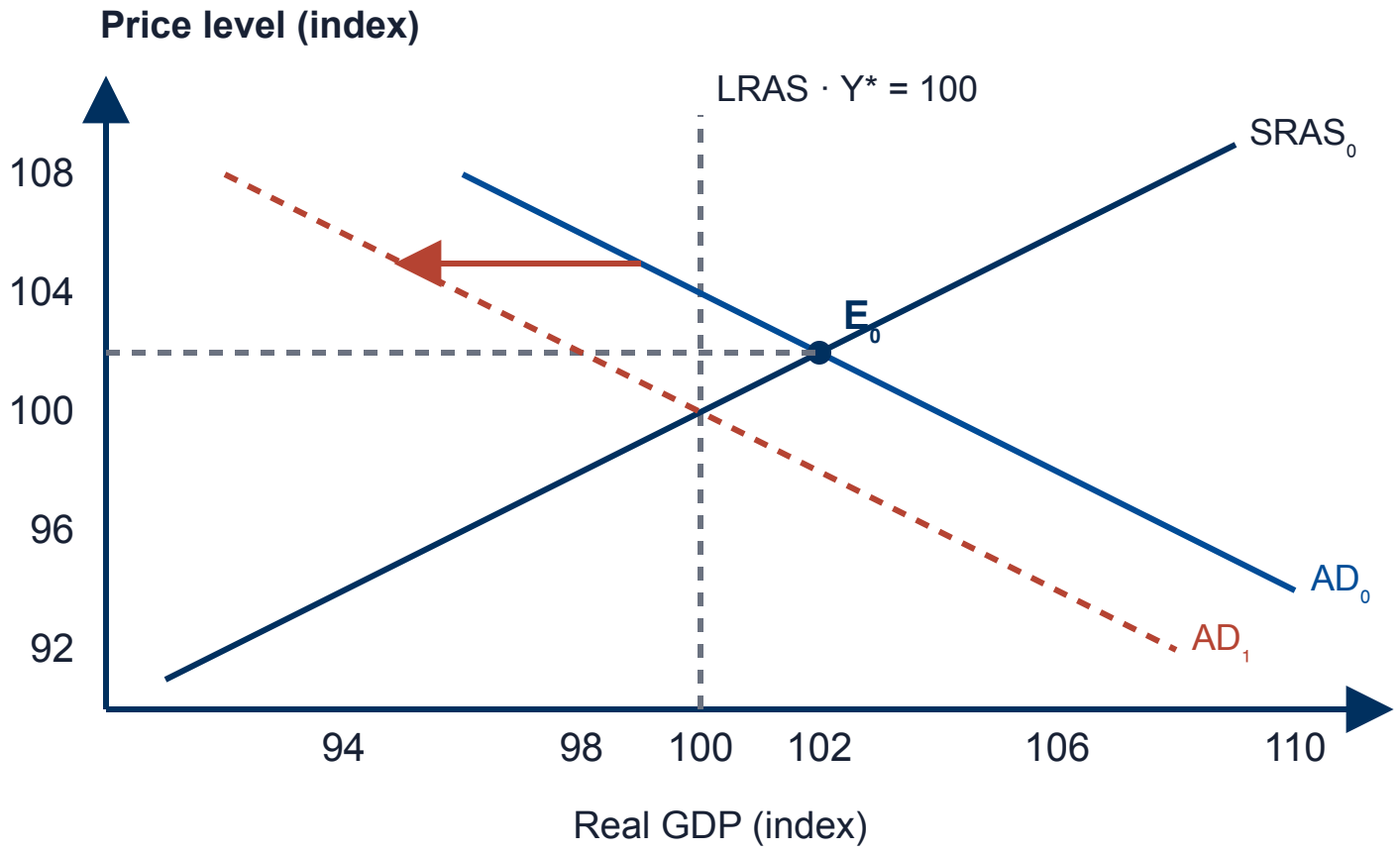
Contractionary Policy and Aggregate Demand



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.
2. Tighter financial conditions restrain interest-sensitive C and I.

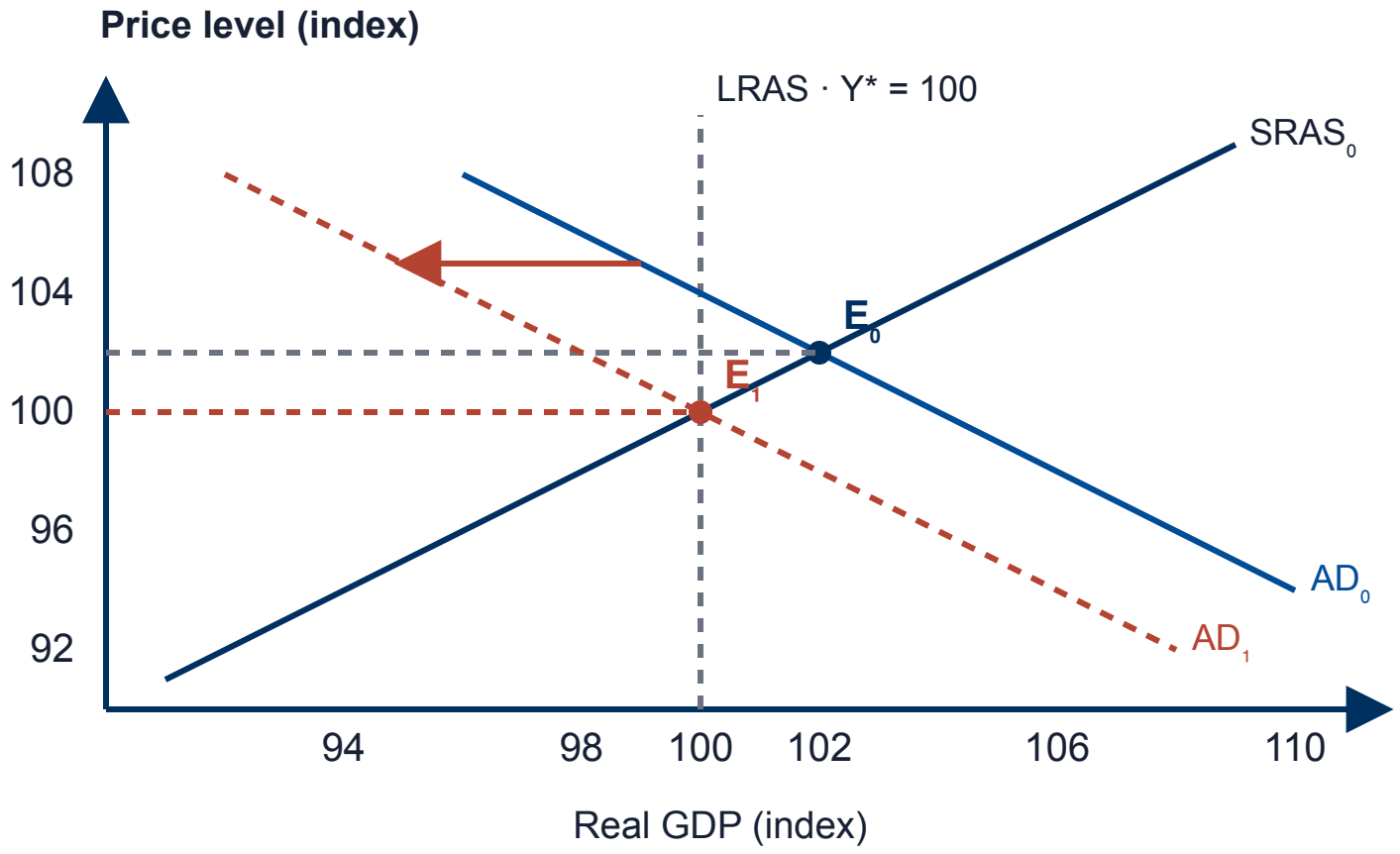
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Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

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1. Begin above potential output: $Y_0 = 102$, while $Y^* = 100$.
2. Tighter financial conditions restrain interest-sensitive C and I.
3. AD shifts left, from AD_0 to AD_1; SRAS and LRAS stay fixed.
4. In this short-run model, output and the price level are lower than without the tightening.

Check the Three Objectives

- A · At a fixed 6% rate, the same bill basket becomes more expensive. Movement or shift in money demand?

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- C · Trace a rate decrease through C and I to AD, output, and the price level.

Explanation. A: MD shifts right. B: the equilibrium rate falls. C: if financial conditions ease, C and I rise, AD shifts right, and short-run output and the price level rise with SRAS fixed.